

Algebraic Geometry 9

Lecture notes with examples

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Solutions: Generic Points, Dimension, Codimension, and Counterexamples

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Scheme Theory: Problems 4–7

Divisor Class Groups, Morphisms of Projective Spaces, Coherent Sheaves,
and Closed Subschemes

Contents

4 Divisor Class Groups of Three Schemes

Problem statement

For each of the following schemes, show the claimed isomorphism. In each case, remove a suitable closed subset.

(a)

$$\mathrm{Cl}(\mathbb{P}_k^1 \times_{\mathrm{Spec} k} \mathbb{P}_k^1) \cong \mathbb{Z}^2.$$

(b) Let

$$X = \mathrm{Spec} k[x, y, z]/(xy - z^2).$$

Show that

$$\mathrm{Cl}(X) \cong \mathbb{Z}/2\mathbb{Z}.$$

(c) Let

$$X = \mathrm{Spec} k[x, y, z, w]/(xy - zw).$$

Show that

$$\mathrm{Cl}(X) \cong \mathbb{Z}.$$

4.1 A localization principle for divisor class groups

We shall repeatedly use the following standard fact.

Proposition 4.1 (Removing codimension-one closed subsets). *Let X be a normal integral Noetherian scheme, and let*

$$U = X \setminus \bigcup_{i=1}^r D_i,$$

where $D_1, \dots, D_r$ are the irreducible codimension-one components of the complement $X \setminus U$. Then there is an exact sequence

$$\bigoplus_{i=1}^r \mathbb{Z}[D_i] \longrightarrow \mathrm{Cl}(X) \longrightarrow \mathrm{Cl}(U) \longrightarrow 0.$$

Consequently, if $\mathrm{Cl}(U) = 0$, then the divisor classes

$$[D_1], \dots, [D_r]$$

generate $\mathrm{Cl}(X)$.

Proof. Every Weil divisor on U is obtained by restricting a Weil divisor on X . Two divisors on X restrict to the same divisor on U precisely when their difference is supported on the codimension-one components of $X \setminus U$. Passing to divisor classes gives the asserted exact sequence. $\square$

The strategy in all three parts is therefore:

1. Remove suitable divisors so that the remaining open subset is factorial.
2. Conclude that the removed prime divisors generate the class group.
3. Compute the relations among their divisor classes using principal divisors.

4.2 $\mathrm{Cl}(\mathbb{P}_k^1 \times \mathbb{P}_k^1) \cong \mathbb{Z}^2$

Solution.

Let

$$X = \mathbb{P}_k^1 \times_{\mathrm{Spec} k} \mathbb{P}_k^1.$$

Choose the two divisors

$$D_1 = \{\infty\} \times \mathbb{P}_k^1, \quad D_2 = \mathbb{P}_k^1 \times \{\infty\}.$$

Their complement is

$$U = X \setminus (D_1 \cup D_2).$$

Since

$$\mathbb{P}_k^1 \setminus \{\infty\} \cong \mathbb{A}_k^1,$$

we obtain

$$U \cong \mathbb{A}_k^1 \times \mathbb{A}_k^1 \cong \mathbb{A}_k^2 = \mathrm{Spec} k[s, t].$$

The polynomial ring $k[s, t]$ is a unique factorization domain. Therefore

$$\mathrm{Cl}(U) = 0.$$

By the localization principle, the classes of the two removed divisors generate the class group:

$$\mathbb{Z}[D_1] \oplus \mathbb{Z}[D_2] \twoheadrightarrow \mathrm{Cl}(X).$$

It remains to show that there is no nontrivial relation between $[D_1]$ and $[D_2]$.

Suppose that

$$aD_1 + bD_2 = \mathrm{div}(f)$$

for some rational function

$$f \in k(X)^\times.$$

Since the divisor of f is supported entirely on $D_1 \cup D_2$, the restriction of f to U has neither zeros nor poles on U . Thus $f|_U$ is a unit in

$$\Gamma(U, \mathcal{O}_U) = k[s, t].$$

But

$$k[s, t]^\times = k^\times.$$

Hence

$$f|_U = c$$

for some nonzero constant $c \in k^\times$. Since U is dense in X , two rational functions agreeing on U agree in the function field. Therefore

$$f = c$$

as a rational function on X . Consequently,

$$\operatorname{div}(f) = 0.$$

Thus

$$aD_1 + bD_2 = 0,$$

and therefore

$$a = b = 0.$$

Hence D_1 and D_2 freely generate the divisor class group. Therefore

$$\boxed{\operatorname{Cl}(\mathbb{P}_k^1 \times_{\operatorname{Spec} k} \mathbb{P}_k^1) \cong \mathbb{Z}[D_1] \oplus \mathbb{Z}[D_2] \cong \mathbb{Z}^2.}$$

Remark 4.2.1. Geometrically, the two generators correspond to the two different rulings of the surface:

$$\{\text{point}\} \times \mathbb{P}_k^1 \quad \text{and} \quad \mathbb{P}_k^1 \times \{\text{point}\}.$$

A divisor of the first type cannot be linearly equivalent to a nonzero multiple of a divisor of the second type.

4.3 The quadric cone $xy = z^2$: $\operatorname{Cl}(X) \cong \mathbb{Z}/2\mathbb{Z}$

Solution.

Let

$$A = k[x, y, z]/(xy - z^2), \quad X = \operatorname{Spec} A.$$

We shall prove that

$$\operatorname{Cl}(X) \cong \mathbb{Z}/2\mathbb{Z}.$$

Normality of X

The scheme X is the affine quadric cone defined by the equation

$$xy = z^2.$$

It is an integral hypersurface. Since hypersurface rings are Cohen–Macaulay, the ring A satisfies Serre’s condition S_2 .

The only singular point is the vertex

$$(x, y, z) = (0, 0, 0).$$

Since X has dimension 2, the singular point has codimension 2. Thus X is regular in codimension 1, so A satisfies Serre’s condition R_1 .

By Serre’s normality criterion, A is normal. Hence the divisor class group $\operatorname{Cl}(X)$ is defined using Weil divisors.

Removing a divisor

Consider the principal open subset

$$U = D(x) \subseteq X.$$

On U , the element x is invertible. From the defining relation

$$xy = z^2$$

we obtain

$$y = \frac{z^2}{x}.$$

Therefore

$$A_x \cong k[x, x^{-1}, z].$$

The ring $k[x, x^{-1}, z]$ is a Laurent polynomial ring over k , hence it is a unique factorization domain. Therefore

$$\text{Cl}(U) = 0.$$

Now compute the complement:

$$X \setminus U = V(x).$$

Inside X ,

$$A/(x) \cong k[y, z]/(z^2).$$

The unique minimal prime of this quotient is generated by z . Therefore $V(x)$ has a single irreducible codimension-one component:

$$D = V(x, z).$$

By the localization principle, $[D]$ generates $\text{Cl}(X)$:

$$\mathbb{Z}[D] \twoheadrightarrow \text{Cl}(X).$$

The relation $2[D] = 0$

We now compute the principal divisor of x .

Let

$$\mathfrak{p} = (x, z) \subseteq A$$

be the height-one prime corresponding to D . At the generic point of D , the element y does not belong to $\mathfrak{p}$, and is therefore invertible in $A_{\mathfrak{p}}$.

From

$$xy = z^2$$

we obtain in the local ring $A_{\mathfrak{p}}$:

$$x = \frac{z^2}{y}.$$

Since y is a unit and z generates the maximal ideal of the discrete valuation ring $A_{\mathfrak{p}}$, we have

$$\text{ord}_D(x) = 2.$$

Thus

$$\text{div}(x) = 2D.$$

Since principal divisors represent zero in the class group,

$$2[D] = 0$$

in $\text{Cl}(X)$.

There are no additional relations

Suppose that

$$nD = \text{div}(f)$$

for some rational function

$$f \in k(X)^\times.$$

Since $\text{div}(f)$ is supported on $X \setminus U$, the restriction $f|_U$ is a unit in

$$A_x \cong k[x, x^{-1}, z].$$

The units of this Laurent polynomial ring are precisely

$$A_x^\times = k^\times x^{\mathbb{Z}}.$$

Hence there exist

$$c \in k^\times, \quad m \in \mathbb{Z}$$

such that

$$f|_U = cx^m.$$

Since U is dense in X , equality on U implies equality in the function field:

$$f = cx^m.$$

Therefore

$$\text{div}(f) = m \text{div}(x) = 2mD.$$

Thus

$$n = 2m.$$

Hence every relation on $[D]$ is generated by the relation $2[D] = 0$. Therefore

$$\boxed{\text{Cl}\left(\text{Spec } k[x, y, z]/(xy - z^2)\right) \cong \mathbb{Z}[D]/(2[D]) \cong \mathbb{Z}/2\mathbb{Z}.$$

The nonzero class is represented by the ruling line

$$D = V(x, z).$$

4.4 The quadric cone $xy = zw$: $\text{Cl}(X) \cong \mathbb{Z}$

Solution.

Let

$$A = k[x, y, z, w]/(xy - zw), \quad X = \text{Spec } A.$$

We shall prove that

$$\text{Cl}(X) \cong \mathbb{Z}.$$

Normality of X

The scheme X is the affine three-dimensional quadric cone defined by

$$xy = zw.$$

Again A is a hypersurface ring, hence it satisfies Serre's condition S_2 .

Its only singular point is the origin

$$(x, y, z, w) = (0, 0, 0, 0).$$

Since X has dimension 3, the singular locus has codimension 3. In particular, X is regular in codimension 1. Thus A satisfies R_1 .

By Serre's normality criterion, A is normal.

Removing a divisor

Consider

$$U = D(x) \subseteq X.$$

Since x is invertible on U , the equation

$$xy = zw$$

gives

$$y = \frac{zw}{x}.$$

Therefore

$$A_x \cong k[x, x^{-1}, z, w].$$

This is a Laurent polynomial ring and hence a unique factorization domain. Therefore

$$\text{Cl}(U) = 0.$$

Now

$$X \setminus U = V(x).$$

We compute

$$A/(x) \cong k[y, z, w]/(zw).$$

The ring $k[y, z, w]/(zw)$ has two minimal primes:

$$(z) \quad \text{and} \quad (w).$$

Therefore $V(x)$ has two irreducible codimension-one components:

$$D = V(x, z), \quad E = V(x, w).$$

Because $\text{Cl}(U) = 0$, these two divisor classes generate $\text{Cl}(X)$:

$$\mathbb{Z}[D] \oplus \mathbb{Z}[E] \twoheadrightarrow \text{Cl}(X).$$

The relation coming from $\text{div}(x)$

We compute the order of vanishing of x along D and E .

At the generic point of

$$D = V(x, z),$$

the elements y and w are invertible. Since

$$xy = zw,$$

we have

$$x = \frac{w}{y}z.$$

The factor $\frac{w}{y}$ is a unit at the generic point of D . Therefore

$$\text{ord}_D(x) = 1.$$

Similarly, at the generic point of

$$E = V(x, w),$$

the elements y and z are invertible, and

$$x = \frac{z}{y}w.$$

Hence

$$\text{ord}_E(x) = 1.$$

Therefore

$$\text{div}(x) = D + E.$$

Since principal divisors represent zero in the divisor class group,

$$[D] + [E] = 0.$$

Thus

$$[E] = -[D],$$

and $\text{Cl}(X)$ is generated by the single class $[D]$.

There is no torsion relation

Suppose that

$$aD + bE = \operatorname{div}(f)$$

for some rational function

$$f \in k(X)^\times.$$

Since the divisor of f is supported on $X \setminus U$, its restriction to U is a unit in

$$A_x \cong k[x, x^{-1}, z, w].$$

The units of this ring are

$$A_x^\times = k^\times x^\mathbb{Z}.$$

Hence

$$f|_U = cx^m$$

for some

$$c \in k^\times, \quad m \in \mathbb{Z}.$$

Since U is dense in X ,

$$f = cx^m$$

in the function field. Consequently,

$$\operatorname{div}(f) = m \operatorname{div}(x) = mD + mE.$$

Thus every relation among D and E is an integral multiple of

$$D + E.$$

Therefore

$$\operatorname{Cl}(X) \cong \frac{\mathbb{Z}[D] \oplus \mathbb{Z}[E]}{\mathbb{Z}([D] + [E])}.$$

Since the quotient of $\mathbb{Z}^2$ by the subgroup generated by $(1, 1)$ is isomorphic to $\mathbb{Z}$, we obtain

$$\boxed{\operatorname{Cl}(\operatorname{Spec} k[x, y, z, w]/(xy - zw)) \cong \mathbb{Z}.}$$

A generator may be taken to be

$$[D] = [V(x, z)],$$

while

$$[E] = -[D].$$

5 Morphisms from Projective Space

Problem statement

Let

$$\varphi : \mathbb{P}^n \longrightarrow \mathbb{P}^m$$

be a morphism. Show that either

$$\varphi(\mathbb{P}^n)$$

is a point, or else

$$m \geq n \quad \text{and} \quad \dim \varphi(\mathbb{P}^n) = n.$$

The suggested strategy is first to prove that if $m < n$, then every morphism

$$\mathbb{P}^n \longrightarrow \mathbb{P}^m$$

is constant, and then to finish by using projections from linear subspaces.

5.1 Morphisms to lower-dimensional projective spaces

Lemma 5.1. *Let*

$$\psi : \mathbb{P}^n \longrightarrow \mathbb{P}^r$$

be a morphism with

$$r < n.$$

Then ψ is constant.

Proof. Since

$$\text{Pic}(\mathbb{P}^n) \cong \mathbb{Z},$$

there exists an integer $d \geq 0$ such that

$$\psi^* \mathcal{O}_{\mathbb{P}^r}(1) \cong \mathcal{O}_{\mathbb{P}^n}(d).$$

We distinguish two cases.

Case 1: $d = 0$

If

$$\psi^* \mathcal{O}_{\mathbb{P}^r}(1) \cong \mathcal{O}_{\mathbb{P}^n},$$

then the pullbacks of the homogeneous coordinate sections on $\mathbb{P}^r$ are global sections of the trivial line bundle on $\mathbb{P}^n$.

But

$$\Gamma(\mathbb{P}^n, \mathcal{O}_{\mathbb{P}^n}) = k.$$

Thus these pullback sections are constants. Hence all points of $\mathbb{P}^n$ are sent to the same point of $\mathbb{P}^r$, and ψ is constant.

Case 2: $d > 0$

If $d > 0$, then ψ is represented by $r + 1$ homogeneous polynomials of the same degree d :

$$F_0, \dots, F_r \in k[X_0, \dots, X_n],$$

with no common zero in $\mathbb{P}^n$.

Indeed,

$$\psi([X_0 : \dots : X_n]) = [F_0(X) : \dots : F_r(X)].$$

Since $r < n$, we have

$$r + 1 \leq n.$$

Consider the homogeneous ideal

$$I = (F_0, \dots, F_r) \subseteq k[X_0, \dots, X_n].$$

By Krull's height theorem,

$$\text{ht}(I) \leq r + 1 \leq n.$$

On the other hand, if the homogeneous polynomials $F_0, \dots, F_r$ had no common zero in projective space, then the radical of I would contain the irrelevant ideal

$$(X_0, \dots, X_n),$$

whose height is

$$n + 1.$$

That would force

$$\text{ht}(I) = n + 1,$$

contradicting

$$\text{ht}(I) \leq n.$$

Therefore the polynomials $F_0, \dots, F_r$ must have a common zero in $\mathbb{P}^n$. But then they do not define a morphism on all of $\mathbb{P}^n$, a contradiction.

Hence the case $d > 0$ is impossible. Therefore $d = 0$, and ψ is constant. $\square$

Thus we have proved:

$$\boxed{r < n \implies \text{every morphism } \mathbb{P}^n \rightarrow \mathbb{P}^r \text{ is constant.}}$$

5.2 Dimension of the image of a nonconstant morphism

Solution.

Let

$$Y = \varphi(\mathbb{P}^n) \subseteq \mathbb{P}^m.$$

Since $\mathbb{P}^n$ is irreducible and φ is continuous, Y is irreducible. Put

$$r = \dim Y.$$

Because Y is the image of an n -dimensional variety,

$$r \leq n.$$

We shall show that if Y is not a point, then in fact

$$r = n.$$

Suppose, for contradiction, that

$$0 < r < n.$$

After extending the base field if necessary, we may choose a sufficiently general linear subspace

$$\Lambda \subseteq \mathbb{P}^m$$

of dimension

$$m - r - 1$$

such that

$$\Lambda \cap Y = \emptyset.$$

Projection away from Λ defines a morphism

$$\pi_\Lambda : Y \longrightarrow \mathbb{P}^r.$$

For a sufficiently general choice of Λ , this linear projection is finite onto its image. Since

$$\dim Y = r > 0,$$

the morphism π_Λ is nonconstant.

Now consider the composite morphism

$$\mathbb{P}^n \xrightarrow{\varphi} Y \xrightarrow{\pi_\Lambda} \mathbb{P}^r.$$

This is a nonconstant morphism

$$\mathbb{P}^n \longrightarrow \mathbb{P}^r.$$

But

$$r < n,$$

so by the lemma every such morphism must be constant. This is a contradiction.

Therefore no intermediate dimension

$$0 < r < n$$

is possible. Hence

$$r = 0 \quad \text{or} \quad r = n.$$

If

$$r = 0,$$

then the irreducible projective variety Y is a single point. Thus

$$\varphi(\mathbb{P}^n)$$

is a point.

If

$$r = n,$$

then $Y \subseteq \mathbb{P}^m$ has dimension n . Since every closed subset of $\mathbb{P}^m$ has dimension at most m , we must have

$$n \leq m.$$

Therefore

either $\varphi(\mathbb{P}^n)$ is a point, or else $m \geq n$ and $\dim \varphi(\mathbb{P}^n) = n$.
--

Remark 5.2.1. The result says that projective space cannot be mapped nontrivially onto a strictly lower-dimensional projective variety by a morphism. A nonconstant morphism from $\mathbb{P}^n$ must preserve the full dimension of $\mathbb{P}^n$.

6 Quasi-Coherent and Coherent Sheaves

Problem statement

Let X be a scheme and let

$$f : \mathcal{F} \longrightarrow \mathcal{G}$$

be a morphism of quasi-coherent sheaves of $\mathcal{O}_X$ -modules.

(a) Show that

$$\ker f, \quad \operatorname{coker} f, \quad \operatorname{im} f$$

are quasi-coherent.

(b) Further, if X is Noetherian and $\mathcal{F}$ and $\mathcal{G}$ are coherent, show that

$$\ker f, \quad \operatorname{coker} f, \quad \operatorname{im} f$$

are coherent.

(c) Let

$$g : X \longrightarrow Y$$

be a morphism of schemes, and let $\mathcal{F}$ be a quasi-coherent, respectively coherent, sheaf of $\mathcal{O}_Y$ -modules. Show that

$$g^* \mathcal{F}$$

is quasi-coherent, respectively coherent, on X .

(d) Show by example that if $\mathcal{G}$ is a coherent sheaf on X , then

$$g_* \mathcal{G}$$

need not be coherent on Y .

6.1 Kernels, cokernels, and images of quasi-coherent sheaves

Solution.

Let

$$f : \mathcal{F} \longrightarrow \mathcal{G}$$

be a morphism of quasi-coherent $\mathcal{O}_X$ -modules.

Take an affine open subset

$$U = \operatorname{Spec} A \subseteq X.$$

Because $\mathcal{F}$ and $\mathcal{G}$ are quasi-coherent, there exist A -modules M and N such that

$$\mathcal{F}|_U \cong \widetilde{M}, \quad \mathcal{G}|_U \cong \widetilde{N}.$$

Since the association

$$M \longmapsto \widetilde{M}$$

gives an equivalence between A -modules and quasi-coherent sheaves on $\operatorname{Spec} A$, the restricted morphism

$$f|_U : \mathcal{F}|_U \longrightarrow \mathcal{G}|_U$$

corresponds to an A -module homomorphism

$$\phi : M \longrightarrow N.$$

For modules on an affine scheme, localization is exact. Therefore sheafification preserves kernels, cokernels, and images:

$$\ker(\widetilde{\phi}) \cong \widetilde{\ker \phi},$$

$$\operatorname{coker}(\widetilde{\phi}) \cong \widetilde{\operatorname{coker} \phi},$$

and

$$\operatorname{im}(\widetilde{\phi}) \cong \widetilde{\operatorname{im} \phi}.$$

Hence

$$(\ker f)|_U \cong \widetilde{\ker \phi},$$

$$(\operatorname{coker} f)|_U \cong \widetilde{\operatorname{coker} \phi},$$

and

$$(\operatorname{im} f)|_U \cong \widetilde{\operatorname{im} \phi}.$$

Each of these is quasi-coherent on the affine open subset U .

Since quasi-coherence is a local property and the affine open subsets cover X , we conclude that

$\ker f, \quad \operatorname{coker} f, \quad \operatorname{im} f$ are quasi-coherent sheaves on X .

6.2 The coherent case on a Noetherian scheme

Solution.

Now suppose that X is Noetherian and that

$$\mathcal{F}, \quad \mathcal{G}$$

are coherent sheaves.

Take an affine open subset

$$U = \operatorname{Spec} A \subseteq X.$$

Since X is Noetherian, the ring A is Noetherian. Since $\mathcal{F}$ and $\mathcal{G}$ are coherent, there exist finitely generated A -modules M and N such that

$$\mathcal{F}|_U \cong \widetilde{M}, \quad \mathcal{G}|_U \cong \widetilde{N}.$$

The restricted morphism is induced by an A -module homomorphism

$$\phi : M \longrightarrow N.$$

Because A is Noetherian:

- every submodule of the finitely generated module M is finitely generated;
- every submodule of the finitely generated module N is finitely generated;
- every quotient of a finitely generated module is finitely generated.

Therefore

$$\ker \phi, \quad \operatorname{im} \phi, \quad \operatorname{coker} \phi$$

are finitely generated A -modules.

Consequently,

$$\widetilde{\ker \phi}, \quad \widetilde{\operatorname{im} \phi}, \quad \widetilde{\operatorname{coker} \phi}$$

are coherent sheaves on U .

Since coherence is local, we conclude that

$\ker f, \quad \operatorname{coker} f, \quad \operatorname{im} f$ are coherent sheaves on X .

Remark 6.2.1. The Noetherian hypothesis is essential for kernels and images. Over a non-Noetherian ring, a submodule of a finitely generated module need not be finitely generated.

6.3 Pullbacks of quasi-coherent and coherent sheaves

Solution.

Let

$$g : X \longrightarrow Y$$

be a morphism of schemes, and let $\mathcal{F}$ be a quasi-coherent sheaf of $\mathcal{O}_Y$ -modules.

Recall that

$$g^*\mathcal{F} = g^{-1}\mathcal{F} \otimes_{g^{-1}\mathcal{O}_Y} \mathcal{O}_X.$$

Take an affine open subset

$$V = \operatorname{Spec} A \subseteq Y.$$

Since $\mathcal{F}$ is quasi-coherent, there exists an A -module M such that

$$\mathcal{F}|_V \cong \widetilde{M}.$$

Now let

$$U = \operatorname{Spec} B \subseteq g^{-1}(V)$$

be an affine open subset of X . The restricted morphism

$$g|_U : U \longrightarrow V$$

corresponds to a ring homomorphism

$$A \longrightarrow B.$$

The pullback of $\widetilde{M}$ to U is given by

$$(g^*\mathcal{F})|_U \cong \widetilde{M \otimes_A B}.$$

Thus, on every affine open subset $U \subseteq X$, the sheaf $g^*\mathcal{F}$ is associated to a module. Hence

$$\boxed{\mathcal{F} \text{ quasi-coherent on } Y \implies g^*\mathcal{F} \text{ quasi-coherent on } X.}$$

Now suppose that $\mathcal{F}$ is coherent.

Locally on $V = \operatorname{Spec} A$, the module M may be taken to be finitely presented. Thus there exists an exact sequence

$$A^p \longrightarrow A^q \longrightarrow M \longrightarrow 0$$

with $p, q < \infty$.

Tensoring with B gives an exact sequence

$$B^p \longrightarrow B^q \longrightarrow M \otimes_A B \longrightarrow 0.$$

Therefore

$$M \otimes_A B$$

is finitely presented as a B -module. Hence

$$\widetilde{M \otimes_A B}$$

is coherent on U .

Since this holds locally on X , we conclude that

$$\boxed{\mathcal{F} \text{ coherent on } Y \implies g^*\mathcal{F} \text{ coherent on } X.}$$

6.4 A pushforward of a coherent sheaf that is not coherent

Solution.

Consider the structure morphism

$$p : \mathbb{A}_k^1 \longrightarrow \operatorname{Spec} k.$$

Let

$$\mathcal{G} = \mathcal{O}_{\mathbb{A}_k^1}.$$

Since $\mathbb{A}_k^1$ is affine and Noetherian, its structure sheaf is coherent.

We now compute the pushforward. Since $\operatorname{Spec} k$ consists of a single point,

$$p_*\mathcal{G}$$

corresponds to the k -module

$$\Gamma(\mathbb{A}_k^1, \mathcal{O}_{\mathbb{A}_k^1}).$$

But

$$\Gamma(\mathbb{A}_k^1, \mathcal{O}_{\mathbb{A}_k^1}) = k[t].$$

As a vector space over k , the ring $k[t]$ has basis

$$1, t, t^2, t^3, \dots$$

and is therefore infinite-dimensional.

A coherent sheaf on $\operatorname{Spec} k$ corresponds to a finitely generated k -module, equivalently, to a finite-dimensional k -vector space. Since $k[t]$ is not finite-dimensional over k , the sheaf

$$p_*\mathcal{O}_{\mathbb{A}_k^1}$$

is not coherent.

Thus

$\mathcal{O}_{\mathbb{A}_k^1}$ is coherent on $\mathbb{A}_k^1$,	$p_*\mathcal{O}_{\mathbb{A}_k^1}$ is not coherent on $\operatorname{Spec} k$.
--	--

In particular, the pushforward of a coherent sheaf need not be coherent.

Remark 6.4.1. In this example, the pushforward is nevertheless quasi-coherent:

$$p_*\mathcal{O}_{\mathbb{A}_k^1} \cong \widetilde{k[t]}.$$

The failure is not quasi-coherence, but finite generation over k .

7 Closed Subschemes and Quasi-Coherent Ideal Sheaves

Problem statement

Let

$$i : Z \hookrightarrow X$$

be a closed immersion of schemes. Recall that this means that i is a homeomorphism of Z onto a closed subset of X , and that the morphism of sheaves

$$i^\# : \mathcal{O}_X \longrightarrow i_* \mathcal{O}_Z$$

is surjective.

Define

$$\mathcal{I}_{Z/X} = \ker i^\#.$$

Show the following.

- (a) $\mathcal{I}_{Z/X}$ is a sheaf of ideals of $\mathcal{O}_X$; that is,

$$\mathcal{I}_{Z/X}(U)$$

is an ideal of $\mathcal{O}_X(U)$ for every open subset $U \subseteq X$.

- (b) $\mathcal{I}_{Z/X}$ is a quasi-coherent sheaf of $\mathcal{O}_X$ -modules, and it is coherent if X is Noetherian.
(c) There is a one-to-one correspondence between quasi-coherent sheaves of ideals on X and closed subschemes of X .

7.1 $\mathcal{I}_{Z/X}$ is a sheaf of ideals

Solution.

By definition,

$$\mathcal{I}_{Z/X} = \ker(\mathcal{O}_X \longrightarrow i_* \mathcal{O}_Z).$$

For every open subset

$$U \subseteq X,$$

we have

$$(i_* \mathcal{O}_Z)(U) = \mathcal{O}_Z(i^{-1}(U)).$$

Since i identifies Z homeomorphically with a closed subset of X , we may write

$$i^{-1}(U) = U \cap Z.$$

Therefore

$$\mathcal{I}_{Z/X}(U) = \ker(\mathcal{O}_X(U) \longrightarrow \mathcal{O}_Z(U \cap Z)).$$

The map

$$\mathcal{O}_X(U) \longrightarrow \mathcal{O}_Z(U \cap Z)$$

is a ring homomorphism. The kernel of a ring homomorphism is an ideal. Hence

$$\mathcal{I}_{Z/X}(U)$$

is an ideal in

$$\mathcal{O}_X(U).$$

Now let

$$V \subseteq U$$

be open subsets of X . The restriction maps fit into a commutative diagram

$$\begin{array}{ccc} \mathcal{O}_X(U) & \longrightarrow & \mathcal{O}_Z(U \cap Z) \\ \downarrow & & \downarrow \\ \mathcal{O}_X(V) & \longrightarrow & \mathcal{O}_Z(V \cap Z). \end{array}$$

Therefore restriction maps send elements of

$$\mathcal{I}_{Z/X}(U)$$

to elements of

$$\mathcal{I}_{Z/X}(V).$$

Thus $\mathcal{I}_{Z/X}$ is a subsheaf of rings of $\mathcal{O}_X$, and on every open set its sections form an ideal. Hence

$$\boxed{\mathcal{I}_{Z/X} \text{ is a sheaf of ideals of } \mathcal{O}_X.}$$

7.2 $\mathcal{I}_{Z/X}$ is quasi-coherent and, in the Noetherian case, coherent

Solution.

Take an affine open subset

$$U = \operatorname{Spec} A \subseteq X.$$

Because

$$i : Z \hookrightarrow X$$

is a closed immersion, the intersection

$$Z \cap U$$

is a closed subscheme of U . Closed subschemes of an affine scheme correspond to ideals. Therefore there exists an ideal

$$I \subseteq A$$

such that

$$Z \cap U \cong \operatorname{Spec}(A/I).$$

On U , the morphism

$$\mathcal{O}_X \longrightarrow i_* \mathcal{O}_Z$$

is induced by the quotient map

$$A \longrightarrow A/I.$$

Its kernel is the ideal I . Passing to associated sheaves, we obtain

$$\mathcal{I}_{Z/X}|_U \cong \tilde{I}.$$

Thus, on every affine open subset $U = \operatorname{Spec} A \subseteq X$, the ideal sheaf $\mathcal{I}_{Z/X}$ is associated to an A -module. Hence it is quasi-coherent:

$$\boxed{\mathcal{I}_{Z/X} \text{ is a quasi-coherent sheaf of } \mathcal{O}_X\text{-modules.}}$$

Now assume that X is Noetherian.

Then for every affine open subset

$$U = \operatorname{Spec} A \subseteq X,$$

the ring A is Noetherian. In particular, every ideal

$$I \subseteq A$$

is finitely generated.

Therefore

$$\tilde{I}$$

is a coherent sheaf on U .

Since coherence is local, it follows that

$$\boxed{X \text{ Noetherian} \implies \mathcal{I}_{Z/X} \text{ is coherent.}}$$

7.3 Correspondence between closed subschemes and quasi-coherent ideal sheaves

Solution.

We shall prove the bijection

$$\boxed{\left\{ \begin{array}{c} \text{closed subschemes} \\ Z \hookrightarrow X \end{array} \right\} \longleftrightarrow \left\{ \begin{array}{c} \text{quasi-coherent sheaves of ideals} \\ \mathcal{I} \subseteq \mathcal{O}_X \end{array} \right\} .}$$

From a closed subscheme to a quasi-coherent ideal sheaf

Let

$$i : Z \hookrightarrow X$$

be a closed immersion.

Define

$$\mathcal{I}_{Z/X} = \ker(\mathcal{O}_X \longrightarrow i_*\mathcal{O}_Z) .$$

By the previous parts:

- $\mathcal{I}_{Z/X}$ is a sheaf of ideals of $\mathcal{O}_X$;
- $\mathcal{I}_{Z/X}$ is quasi-coherent.

Therefore every closed subscheme determines a quasi-coherent ideal sheaf.

From a quasi-coherent ideal sheaf to a closed subscheme

Conversely, let

$$\mathcal{I} \subseteq \mathcal{O}_X$$

be a quasi-coherent sheaf of ideals.

Take an affine open subset

$$U = \operatorname{Spec} A \subseteq X.$$

Since $\mathcal{I}$ is quasi-coherent, there exists an ideal

$$I_U \subseteq A$$

such that

$$\mathcal{I}|_U \cong \widetilde{I_U}.$$

Define a closed subscheme of U by

$$Z_U = \operatorname{Spec}(A/I_U).$$

Its underlying closed subset is

$$V(I_U) \subseteq \operatorname{Spec} A.$$

Now suppose

$$V = \operatorname{Spec} B \subseteq U$$

is another affine open subset. Since the ideal sheaf restricts compatibly, the ideal defining Z_V is the localization of I_U corresponding to V . Consequently,

$$Z_V$$

is precisely the restriction of

$$Z_U$$

to V .

Thus the affine closed subschemes

$$Z_U = \operatorname{Spec}(A/I_U)$$

agree on overlaps and glue together to form a scheme Z together with a closed immersion

$$i : Z \hookrightarrow X.$$

Locally, the structure sheaf of Z is

$$\mathcal{O}_Z|_{Z \cap U} \cong \widetilde{A/I_U}.$$

Equivalently, the quotient sheaf

$$\mathcal{O}_X/\mathcal{I}$$

is the sheaf of rings defining the closed subscheme Z .

The morphism

$$\mathcal{O}_X \longrightarrow i_*\mathcal{O}_Z$$

has kernel exactly $\mathcal{I}$. Hence

$$\mathcal{I}_{Z/X} = \mathcal{I}.$$

The constructions are inverse

Starting with a closed subscheme

$$Z \hookrightarrow X,$$

forming its ideal sheaf

$$\mathcal{I}_{Z/X},$$

and then reconstructing the closed subscheme locally by quotient rings gives back the original closed subscheme Z .

Conversely, starting with a quasi-coherent ideal sheaf

$$\mathcal{I} \subseteq \mathcal{O}_X,$$

forming the local closed subschemes

$$\mathrm{Spec}(A/I_U),$$

and then taking the ideal sheaf of the resulting closed immersion gives back $\mathcal{I}$.

Therefore the constructions are mutually inverse, and we obtain the required bijection:

$$\boxed{\{\text{closed subschemes of } X\} \longleftrightarrow \{\text{quasi-coherent ideal sheaves } \mathcal{I} \subseteq \mathcal{O}_X\} .}$$

7.4 The affine special case

When

$$X = \mathrm{Spec} A$$

is affine, the correspondence becomes especially concrete.

Every ideal

$$I \subseteq A$$

defines a closed subscheme

$$\mathrm{Spec}(A/I) \hookrightarrow \mathrm{Spec} A.$$

Conversely, every closed subscheme of $\mathrm{Spec} A$ arises uniquely in this way.

Thus

$$\boxed{\{\text{closed subschemes of } \mathrm{Spec} A\} \longleftrightarrow \{\text{ideals } I \subseteq A\}, \quad I \longmapsto \mathrm{Spec}(A/I) .}$$

This is the scheme-theoretic refinement of the ordinary correspondence between ideals of A and closed subsets of $\mathrm{Spec} A$: different ideals may define the same underlying closed set, but they define different closed subscheme structures.

For example,

$$(x) \subseteq k[x] \quad \text{and} \quad (x^2) \subseteq k[x]$$

both define the same underlying closed point

$$\{(x)\} \subseteq \mathbb{A}_k^1,$$

but the closed subschemes

$$\mathrm{Spec} k[x]/(x) \quad \text{and} \quad \mathrm{Spec} k[x]/(x^2)$$

are different. The second is a nonreduced thickening of the first.

8 Theory Behind Problems 4–7: Divisor Classes, Projective Morphisms, Coherent Sheaves, and Closed Subschemes

The problems above bring together four fundamental themes of scheme theory:

1. divisor class groups measure the failure of codimension-one subvarieties to be principal;
2. morphisms from projective space are strongly constrained by positivity and dimension;
3. quasi-coherent and coherent sheaves translate geometric constructions into module-theoretic constructions on affine charts;
4. closed subschemes are encoded exactly by quasi-coherent ideal sheaves.

Although these topics appear different, they are connected by one central principle:

Global geometric questions on schemes are often solved by passing to affine open subsets, performing algebra there, and then gluing the conclusions globally.

8.1 Divisors and divisor class groups on normal schemes

Codimension-one geometry

Let X be an integral Noetherian scheme. A closed irreducible subset

$$D \subseteq X$$

of codimension one is called a *prime divisor*.

Prime divisors are the scheme-theoretic analogues of irreducible hypersurfaces. On an affine variety, they are usually cut out locally by a single irreducible equation. However, globally a codimension-one subvariety need not be defined by one global function.

This difference is precisely what the divisor class group detects.

A *Weil divisor* on X is a finite formal integral linear combination of prime divisors:

$$D = \sum_{Y \subseteq X, \text{codim}(Y,X)=1} n_Y [Y], \quad n_Y \in \mathbb{Z},$$

where all but finitely many n_Y are zero.

The abelian group of Weil divisors is denoted

$$\text{Div}(X).$$

Thus a divisor records codimension-one vanishing and pole data. For example, if

$$D = 2D_1 - D_2,$$

then D should be interpreted as having multiplicity 2 along D_1 and multiplicity -1 , that is, a pole, along D_2 .

Why normality is important

Suppose that X is normal and integral. Then every local ring

$$\mathcal{O}_{X, \eta_D}$$

at the generic point η_D of a prime divisor D is a discrete valuation ring.

Consequently, every nonzero rational function

$$f \in K(X)^\times$$

has a well-defined order of vanishing along D :

$$\text{ord}_D(f) \in \mathbb{Z}.$$

The integer $\text{ord}_D(f)$ has the following meaning:

$$\begin{aligned} \text{ord}_D(f) > 0 &\iff f \text{ vanishes along } D, \\ \text{ord}_D(f) < 0 &\iff f \text{ has a pole along } D, \end{aligned}$$

and

$$\text{ord}_D(f) = 0 \iff f \text{ is a unit at the generic point of } D.$$

The *principal divisor* associated to f is

$$\text{div}(f) = \sum_D \text{ord}_D(f)[D].$$

Only finitely many coefficients are nonzero, so this is a Weil divisor.

Definition 8.1.1. Let X be a normal integral Noetherian scheme. The *divisor class group* of X is

$$\text{Cl}(X) = \frac{\text{Div}(X)}{\{\text{div}(f) : f \in K(X)^\times\}}.$$

Thus two divisors represent the same class precisely when their difference is the divisor of a rational function.

A divisor class is zero if and only if the divisor is principal.

Geometric meaning of the class group

The group $\text{Cl}(X)$ measures the obstruction to describing codimension-one geometry by global rational functions.

If

$$\text{Cl}(X) = 0,$$

then every Weil divisor is principal. In an affine normal setting this is closely related to unique factorization.

For a normal domain A ,

$$\text{Cl}(\text{Spec } A) = 0$$

if A is a unique factorization domain. Indeed, height-one prime ideals correspond to irreducible codimension-one subvarieties, and unique factorization forces each such divisor to arise from one equation.

Thus:

$$\boxed{A \text{ a normal UFD} \implies \text{Cl}(\text{Spec } A) = 0.}$$

This explains why polynomial rings and Laurent polynomial rings are so useful in class-group calculations:

$$k[x_1, \dots, x_n], \quad k[x_1^{\pm 1}, x_2, \dots, x_n]$$

are UFDs, so their spectra have trivial class group.

Removing divisors and the localization sequence

A powerful method for computing divisor class groups is to remove a closed subset until the remaining open subscheme becomes factorial.

Let X be normal and integral, and let

$$U = X \setminus \bigcup_{i=1}^r D_i,$$

where the D_i are precisely the irreducible codimension-one components of $X \setminus U$.

Then restriction of divisors gives an exact sequence

$$\bigoplus_{i=1}^r \mathbb{Z}[D_i] \longrightarrow \text{Cl}(X) \longrightarrow \text{Cl}(U) \longrightarrow 0.$$

The meaning of this sequence is simple:

- divisors on X restrict to divisors on U ;
- divisors supported entirely in $X \setminus U$ disappear after restriction;
- the only codimension-one divisors that disappear are the removed divisors D_i .

If U is factorial, then

$$\text{Cl}(U) = 0.$$

Hence the removed divisors generate $\text{Cl}(X)$:

$$\text{Cl}(X) = \langle [D_1], \dots, [D_r] \rangle.$$

The remaining question is to find all relations among those generators. These relations arise from principal divisors of rational functions whose zero and pole loci lie in the removed boundary.

Units on the factorial open subset

Suppose

$$\operatorname{div}(f)$$

is supported entirely in $X \setminus U$. Then on U , the rational function f has no zeros and no poles. Therefore

$$f|_U \in \Gamma(U, \mathcal{O}_U)^\times.$$

This converts a geometric relation in $\operatorname{Cl}(X)$ into a computation of units in a ring.

For example:

$$k[s, t]^\times = k^\times,$$

whereas

$$k[x, x^{-1}, z]^\times = k^\times x^\mathbb{Z}.$$

These two unit computations lead to two different types of behavior:

- if the only units are constants, then there are no nontrivial divisor relations supported on the boundary;
- if a Laurent monomial such as x^m is also a unit, then powers of x produce relations among boundary divisors.

This is the basic mechanism behind the three computations in Problem 4.

8.2 The class group of a product of projective lines

The surface

$$\mathbb{P}_k^1 \times \mathbb{P}_k^1$$

is the simplest example of a projective surface carrying two independent families of curves.

The two projections

$$\pi_1 : \mathbb{P}_k^1 \times \mathbb{P}_k^1 \longrightarrow \mathbb{P}_k^1, \quad \pi_2 : \mathbb{P}_k^1 \times \mathbb{P}_k^1 \longrightarrow \mathbb{P}_k^1$$

produce two natural divisor classes:

$$F_1 = \{\text{point}\} \times \mathbb{P}_k^1, \quad F_2 = \mathbb{P}_k^1 \times \{\text{point}\}.$$

These are called the two *rulings* of the surface.

Removing one representative from each ruling gives

$$(\mathbb{P}_k^1 \times \mathbb{P}_k^1) \setminus ((\{\infty\} \times \mathbb{P}_k^1) \cup (\mathbb{P}_k^1 \times \{\infty\})) \cong \mathbb{A}_k^2.$$

Since

$$\operatorname{Cl}(\mathbb{A}_k^2) = 0,$$

the two boundary divisors generate the class group.

What distinguishes this example from the affine cones is that

$$\Gamma(\mathbb{A}_k^2, \mathcal{O}_{\mathbb{A}_k^2})^\times = k^\times.$$

Thus no nonconstant rational function can have its entire divisor supported only on these two boundary curves. Hence the two ruling classes remain independent:

$$\mathrm{Cl}(\mathbb{P}_k^1 \times \mathbb{P}_k^1) \cong \mathbb{Z}^2.$$

There is also a line-bundle interpretation. Since the surface is smooth, every Weil divisor is Cartier, and therefore

$$\mathrm{Cl}(\mathbb{P}_k^1 \times \mathbb{P}_k^1) \cong \mathrm{Pic}(\mathbb{P}_k^1 \times \mathbb{P}_k^1).$$

The corresponding line bundles are

$$\mathcal{O}_{\mathbb{P}^1 \times \mathbb{P}^1}(a, b) = \pi_1^* \mathcal{O}_{\mathbb{P}^1}(a) \otimes \pi_2^* \mathcal{O}_{\mathbb{P}^1}(b),$$

and every line bundle is of this form. Hence

$$\mathrm{Pic}(\mathbb{P}_k^1 \times \mathbb{P}_k^1) \cong \mathbb{Z} \oplus \mathbb{Z}.$$

The pair (a, b) records the degree of the divisor in the two ruling directions.

8.3 Quadric cones and nontrivial divisor classes

The surface cone $xy = z^2$

Consider

$$X = \mathrm{Spec} k[x, y, z]/(xy - z^2).$$

This is a two-dimensional affine cone. Away from the origin it is smooth, but the vertex creates nontrivial divisor-class behavior.

If we invert x , then

$$y = \frac{z^2}{x},$$

so

$$\mathcal{O}_X(D(x)) \cong k[x, x^{-1}, z].$$

This ring is factorial. Therefore all divisor-class information is concentrated in the closed set

$$V(x).$$

Scheme-theoretically,

$$A/(x) \cong k[y, z]/(z^2).$$

The underlying irreducible codimension-one subset is

$$D = V(x, z).$$

The important point is that x does not vanish simply along D . At the generic point of D , the element y is a unit and

$$x = \frac{z^2}{y}.$$

Thus x vanishes to order 2 along D :

$$\operatorname{div}(x) = 2D.$$

Hence the class $[D]$ is not necessarily zero, but twice the class is zero:

$$2[D] = 0.$$

This is the geometric origin of the torsion class

$$\operatorname{Cl}(X) \cong \mathbb{Z}/2\mathbb{Z}.$$

The cone has a divisor which behaves like “half” of a principal divisor: the function x defines twice the ruling divisor.

The three-dimensional cone $xy = zw$

Now consider

$$X = \operatorname{Spec} k[x, y, z, w]/(xy - zw).$$

Again inverting x makes the ring factorial:

$$A_x \cong k[x, x^{-1}, z, w].$$

However, the boundary $V(x)$ now has two irreducible codimension-one components:

$$D = V(x, z), \quad E = V(x, w).$$

At the generic point of D ,

$$x = \frac{w}{y}z,$$

so x vanishes to order 1 along D . Similarly, at the generic point of E ,

$$x = \frac{z}{y}w,$$

so x vanishes to order 1 along E .

Thus

$$\operatorname{div}(x) = D + E.$$

This gives the relation

$$[D] + [E] = 0.$$

Unlike the surface cone $xy = z^2$, there is no multiplicity 2 along a single divisor. Instead, one principal divisor splits into two different prime divisors. Therefore one divisor class remains freely generated:

$$\operatorname{Cl}(X) \cong \mathbb{Z}.$$

Conceptually, the comparison is:

Scheme	Boundary divisors	Principal relation
$\text{Spec } k[x, y, z]/(xy - z^2)$	$D = V(x, z)$	$\text{div}(x) = 2D$
$\text{Spec } k[x, y, z, w]/(xy - zw)$	$D = V(x, z), E = V(x, w)$	$\text{div}(x) = D + E$

The first relation produces torsion:

$$\mathbb{Z}/(2) \cong \mathbb{Z}/2\mathbb{Z}.$$

The second relation removes one of two free generators:

$$\mathbb{Z}^2/\mathbb{Z}(1, 1) \cong \mathbb{Z}.$$

8.4 Morphisms from projective space and positivity

Morphisms to projective space as homogeneous polynomials

A morphism

$$\varphi : \mathbb{P}^n \longrightarrow \mathbb{P}^m$$

is controlled by a line bundle and a collection of global sections.

Indeed, let

$$\mathcal{O}_{\mathbb{P}^m}(1)$$

be the hyperplane line bundle on $\mathbb{P}^m$. Its pullback is a line bundle on $\mathbb{P}^n$:

$$\varphi^* \mathcal{O}_{\mathbb{P}^m}(1).$$

Since

$$\text{Pic}(\mathbb{P}^n) \cong \mathbb{Z},$$

every line bundle on $\mathbb{P}^n$ is of the form

$$\mathcal{O}_{\mathbb{P}^n}(d)$$

for a unique integer d . Since $\varphi^* \mathcal{O}_{\mathbb{P}^m}(1)$ is generated by global sections, one necessarily has

$$d \geq 0.$$

Thus

$$\varphi^* \mathcal{O}_{\mathbb{P}^m}(1) \cong \mathcal{O}_{\mathbb{P}^n}(d)$$

for some $d \geq 0$.

The morphism φ is then given by $m + 1$ global sections of

$$\mathcal{O}_{\mathbb{P}^n}(d),$$

that is, by homogeneous polynomials

$$F_0, \dots, F_m$$

of the same degree d , with no common zero:

$$\varphi([X_0 : \dots : X_n]) = [F_0(X) : \dots : F_m(X)].$$

The condition that the F_i have no common zero is essential. If they vanished simultaneously at a point, then the projective coordinate expression would be undefined there.

Constant maps and degree zero

If

$$d = 0,$$

then each F_i is a constant. Therefore the image consists of a single projective point:

$$\varphi(\mathbb{P}^n) = \{[c_0 : \cdots : c_m]\}.$$

Thus:

$$\boxed{\varphi^* \mathcal{O}_{\mathbb{P}^m}(1) \cong \mathcal{O}_{\mathbb{P}^n} \implies \varphi \text{ is constant.}}$$

Conversely, a constant map clearly pulls back $\mathcal{O}_{\mathbb{P}^m}(1)$ to the trivial line bundle.

Hence nonconstant morphisms correspond to the case

$$d > 0.$$

Why there are no nonconstant morphisms $\mathbb{P}^n \rightarrow \mathbb{P}^r$ for $r < n$

Suppose

$$\psi : \mathbb{P}^n \longrightarrow \mathbb{P}^r$$

is nonconstant, where

$$r < n.$$

Then ψ is given by $r + 1$ homogeneous polynomials

$$F_0, \dots, F_r$$

of a positive common degree.

For these polynomials to define a morphism, they must have no common zero in $\mathbb{P}^n$.

However, $r + 1 \leq n$. Geometrically, fewer than $n + 1$ positive-degree hypersurfaces in $\mathbb{P}^n$ cannot cut out the empty set. Their intersection must contain at least one projective point.

Algebraically, the ideal

$$(F_0, \dots, F_r) \subseteq k[X_0, \dots, X_n]$$

has height at most

$$r + 1 \leq n,$$

whereas the irrelevant ideal

$$(X_0, \dots, X_n)$$

has height

$$n + 1.$$

Thus the radical of $(F_0, \dots, F_r)$ cannot contain the irrelevant ideal. Therefore the F_i possess a common projective zero, contradicting the assumption that they define a morphism everywhere.

Hence:

$$\boxed{r < n \implies \text{every morphism } \mathbb{P}^n \longrightarrow \mathbb{P}^r \text{ is constant.}}$$

This is a particularly strong rigidity property of projective space.

Projection from a linear subspace

Suppose now that

$$\varphi : \mathbb{P}^n \longrightarrow \mathbb{P}^m$$

has image

$$Y = \varphi(\mathbb{P}^n)$$

of dimension

$$r.$$

If

$$0 < r < n,$$

one can choose a sufficiently general linear subspace

$$\Lambda \subseteq \mathbb{P}^m$$

of dimension

$$m - r - 1$$

which does not meet Y .

Projection away from Λ gives a morphism

$$\pi_\Lambda : Y \longrightarrow \mathbb{P}^r.$$

For general Λ , this projection is finite onto its image. In particular, it is nonconstant whenever

$$r > 0.$$

The composite

$$\mathbb{P}^n \xrightarrow{\varphi} Y \xrightarrow{\pi_\Lambda} \mathbb{P}^r$$

would therefore be a nonconstant morphism from $\mathbb{P}^n$ to $\mathbb{P}^r$ with

$$r < n.$$

But such a morphism cannot exist. Consequently, the dimension of the image cannot satisfy

$$0 < \dim \varphi(\mathbb{P}^n) < n.$$

Therefore the image is either zero-dimensional or full-dimensional:

$$\boxed{\dim \varphi(\mathbb{P}^n) = 0 \quad \text{or} \quad \dim \varphi(\mathbb{P}^n) = n.}$$

Since an irreducible zero-dimensional projective variety is a point, this becomes:

$$\boxed{\varphi(\mathbb{P}^n) \text{ is a point, or } \dim \varphi(\mathbb{P}^n) = n.}$$

In the nonconstant case,

$$\varphi(\mathbb{P}^n) \subseteq \mathbb{P}^m$$

has dimension n , so necessarily

$$m \geq n.$$

8.5 Quasi-coherent sheaves as geometric modules

From modules to sheaves

Let

$$X = \operatorname{Spec} A$$

be an affine scheme and let M be an A -module. Associated to M is a sheaf of $\mathcal{O}_X$ -modules

$$\widetilde{M}.$$

Its defining property on distinguished open subsets is

$$\widetilde{M}(D(f)) = M_f.$$

Thus localization of modules becomes restriction of sheaves.

For example:

$$\widetilde{A} = \mathcal{O}_X,$$

and for an ideal $I \subseteq A$,

$$\widetilde{I} \subseteq \mathcal{O}_X$$

is the corresponding ideal sheaf.

Definition 8.5.1. A sheaf $\mathcal{F}$ of $\mathcal{O}_X$ -modules is *quasi-coherent* if every point has an affine open neighbourhood

$$U = \operatorname{Spec} A$$

such that

$$\mathcal{F}|_U \cong \widetilde{M}$$

for some A -module M .

Quasi-coherent sheaves are therefore the sheaves that locally come from modules.

This makes them the natural class of sheaves in algebraic geometry. They retain enough algebraic structure to be manipulated locally, while being flexible enough to glue across different affine charts.

The affine equivalence

For an affine scheme

$$X = \operatorname{Spec} A,$$

there is an equivalence of categories

$$\{A\text{-modules}\} \longleftrightarrow \{\text{quasi-coherent } \mathcal{O}_X\text{-modules}\},$$

given by

$$M \longmapsto \widetilde{M}$$

and

$$\mathcal{F} \longmapsto \Gamma(X, \mathcal{F}).$$

This means that on affine schemes, questions about quasi-coherent sheaves are exactly questions about modules.

In particular, if

$$\phi : M \longrightarrow N$$

is a module homomorphism, then

$$\widetilde{\phi} : \widetilde{M} \longrightarrow \widetilde{N}$$

is the corresponding sheaf morphism.

Because localization is exact, one has:

$$\ker(\widetilde{\phi}) \cong \widetilde{\ker \phi},$$

$$\operatorname{im}(\widetilde{\phi}) \cong \widetilde{\operatorname{im} \phi},$$

and

$$\operatorname{coker}(\widetilde{\phi}) \cong \widetilde{\operatorname{coker} \phi}.$$

This is the local algebraic reason that kernels, images, and cokernels of morphisms of quasi-coherent sheaves remain quasi-coherent.

Abelian-category behavior

The category of all sheaves of $\mathcal{O}_X$ -modules is an abelian category: kernels, cokernels, and images always exist as sheaves.

The nontrivial statement is that when one begins with quasi-coherent sheaves, these constructions do not leave the quasi-coherent world:

$$\mathcal{F}, \mathcal{G} \text{ quasi-coherent} \implies \ker f, \operatorname{im} f, \operatorname{coker} f \text{ quasi-coherent.}$$

Therefore quasi-coherent sheaves form an abelian subcategory of the category of all $\mathcal{O}_X$ -modules.

This is fundamental because exact sequences such as

$$0 \longrightarrow \mathcal{F}' \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}'' \longrightarrow 0$$

can be studied locally as exact sequences of modules:

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0.$$

8.6 Coherent sheaves and finiteness

Finite generation and finite presentation

Quasi-coherent sheaves may be extremely large. For instance, on

$$\operatorname{Spec} k$$

every k -vector space defines a quasi-coherent sheaf, including infinite-dimensional ones.

Coherent sheaves are the finite objects among quasi-coherent sheaves.

On a locally Noetherian scheme, a quasi-coherent sheaf $\mathcal{F}$ is coherent precisely when, on every affine open subset

$$U = \operatorname{Spec} A,$$

one has

$$\mathcal{F}|_U \cong \widetilde{M}$$

for a finitely generated A -module M .

Since A is Noetherian, finitely generated modules have finitely generated submodules. Therefore kernels and images remain finitely generated.

This is exactly the point at which the Noetherian hypothesis becomes important.

Why Noetherian schemes preserve coherence under kernels

Suppose

$$f : \mathcal{F} \longrightarrow \mathcal{G}$$

is a morphism of coherent sheaves on a Noetherian scheme X .

Locally, on

$$U = \operatorname{Spec} A,$$

this becomes a homomorphism

$$\phi : M \longrightarrow N$$

of finitely generated A -modules.

Because A is Noetherian:

$$\ker \phi$$

is finitely generated,

$$\operatorname{im} \phi$$

is finitely generated, and

$$\operatorname{coker} \phi$$

is finitely generated.

Therefore:

$\ker f,$	$\operatorname{im} f,$	$\operatorname{coker} f$ are coherent.
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Without Noetherianity, a submodule of a finitely generated module need not be finitely generated. Consequently, kernels and images of morphisms between finite objects may fail to remain finite.

8.7 Pullback and pushforward of sheaves

Pullback as extension of scalars

Let

$$g : X \longrightarrow Y$$

be a morphism of schemes.

For a sheaf $\mathcal{F}$ of $\mathcal{O}_Y$ -modules, the pullback is defined by

$$g^*\mathcal{F} = g^{-1}\mathcal{F} \otimes_{g^{-1}\mathcal{O}_Y} \mathcal{O}_X.$$

On affine charts, this construction is simply tensor product.

Suppose

$$V = \operatorname{Spec} A \subseteq Y, \quad U = \operatorname{Spec} B \subseteq g^{-1}(V),$$

and the restricted morphism corresponds to a ring map

$$A \longrightarrow B.$$

If

$$\mathcal{F}|_V \cong \widetilde{M},$$

then

$$(g^*\mathcal{F})|_U \cong \widetilde{M \otimes_A B}.$$

Thus pullback is geometrically the same operation as extending scalars:

$$M \rightsquigarrow M \otimes_A B.$$

Since tensor products of modules are modules, pullback preserves quasi-coherence:

$$\boxed{\mathcal{F} \text{ quasi-coherent} \implies g^*\mathcal{F} \text{ quasi-coherent.}}$$

If M has a finite presentation

$$A^p \longrightarrow A^q \longrightarrow M \longrightarrow 0,$$

then tensoring with B gives

$$B^p \longrightarrow B^q \longrightarrow M \otimes_A B \longrightarrow 0.$$

Hence finite presentation is preserved under pullback. In the standard Noetherian setting of coherent sheaves, this implies:

$$\boxed{\mathcal{F} \text{ coherent} \implies g^*\mathcal{F} \text{ coherent.}}$$

Remark 8.7.1. For arbitrary non-Noetherian schemes, coherence must be handled with care, because finite presentation and coherence need not coincide without additional hypotheses. In the usual locally Noetherian setting, the pullback statement above is exactly the required form.

Pushforward and the loss of finiteness

The direct image or pushforward of a sheaf $\mathcal{G}$ on X is defined by

$$(g_*\mathcal{G})(V) = \mathcal{G}(g^{-1}(V))$$

for open subsets

$$V \subseteq Y.$$

Unlike pullback, pushforward does not behave locally as a simple finite module operation. It collects all sections over inverse images of open sets, and these spaces of sections may be very large.

For example, consider

$$p : \mathbb{A}_k^1 \longrightarrow \operatorname{Spec} k.$$

The coherent sheaf

$$\mathcal{O}_{\mathbb{A}_k^1}$$

has pushforward corresponding to

$$\Gamma(\mathbb{A}_k^1, \mathcal{O}_{\mathbb{A}_k^1}) = k[t].$$

As a k -vector space,

$$k[t]$$

is infinite-dimensional. Hence it is not a finite k -module, and the corresponding sheaf on $\operatorname{Spec} k$ is not coherent.

Thus:

pushforward does not preserve coherence in general.

The underlying reason is that coherence expresses local finite generation, while global sections over a nonproper space may form an infinite module.

This also explains why properness is so important in algebraic geometry. A fundamental theorem states that for a proper morphism of Noetherian schemes, direct images of coherent sheaves are coherent. The failure in the affine-line example occurs because

$$\mathbb{A}_k^1 \longrightarrow \operatorname{Spec} k$$

is not proper.

Quasi-coherence of pushforwards

Pushforward is better behaved with respect to quasi-coherence than with respect to coherence. Under standard hypotheses, for example when g is quasi-compact and quasi-separated, one has

$$\mathcal{G} \text{ quasi-coherent} \implies g_*\mathcal{G} \text{ quasi-coherent}.$$

In the affine case this is immediate. If

$$g : \operatorname{Spec} B \longrightarrow \operatorname{Spec} A$$

corresponds to a ring map

$$A \longrightarrow B$$

and

$$\mathcal{G} = \widetilde{N}$$

for a B -module N , then

$$g_*\mathcal{G}$$

is the quasi-coherent sheaf associated to the same abelian group N , now viewed as an A -module by restriction of scalars.

However, even when this pushforward remains quasi-coherent, it need not be coherent, because N may fail to be finitely generated as an A -module.

8.8 Closed immersions and ideal sheaves

Closed subsets versus closed subschemes

A closed subset of an affine scheme

$$X = \operatorname{Spec} A$$

is determined by a radical ideal:

$$V(I) = V(\sqrt{I}).$$

Thus the topological closed subset cannot distinguish between I and $\sqrt{I}$.

A closed *subscheme*, however, retains the entire ideal I , including nilpotent information.

For example, inside

$$\mathbb{A}_k^1 = \operatorname{Spec} k[x],$$

the ideals

$$(x) \quad \text{and} \quad (x^2)$$

have the same vanishing set:

$$V(x) = V(x^2) = \{0\}.$$

But they define different closed subschemes:

$$\operatorname{Spec} k[x]/(x) \quad \text{and} \quad \operatorname{Spec} k[x]/(x^2).$$

The first is an ordinary reduced point. The second is a nonreduced point with an infinitesimal thickening. Indeed, in

$$k[x]/(x^2),$$

the class of x is nonzero but satisfies

$$x^2 = 0.$$

Therefore closed subschemes encode more geometry than closed subsets.

Definition of a closed immersion

A morphism

$$i : Z \longrightarrow X$$

is a *closed immersion* if:

1. the map of topological spaces identifies Z homeomorphically with a closed subset of X ;
2. the induced morphism of sheaves of rings

$$i^\# : \mathcal{O}_X \longrightarrow i_*\mathcal{O}_Z$$

is surjective.

The kernel of this surjection is denoted

$$\mathcal{I}_{Z/X} = \ker(\mathcal{O}_X \longrightarrow i_*\mathcal{O}_Z).$$

There is therefore an exact sequence

$$0 \longrightarrow \mathcal{I}_{Z/X} \longrightarrow \mathcal{O}_X \longrightarrow i_*\mathcal{O}_Z \longrightarrow 0.$$

The sheaf $\mathcal{I}_{Z/X}$ is called the *ideal sheaf* defining Z inside X .

Why the kernel is a sheaf of ideals

For every open subset

$$U \subseteq X,$$

one has

$$\mathcal{I}_{Z/X}(U) = \ker(\mathcal{O}_X(U) \longrightarrow \mathcal{O}_Z(U \cap Z)).$$

Since this is the kernel of a ring homomorphism,

$$\mathcal{I}_{Z/X}(U)$$

is an ideal of

$$\mathcal{O}_X(U).$$

Thus the ideal-sheaf property is simply the sheaf-theoretic version of the elementary algebraic fact that the kernel of a quotient map

$$A \longrightarrow A/I$$

is the ideal I .

The affine local form of a closed immersion

Every closed immersion is affine locally of the form

$$\mathrm{Spec}(A/I) \hookrightarrow \mathrm{Spec} A.$$

Indeed, if

$$U = \mathrm{Spec} A \subseteq X$$

is affine and $Z \hookrightarrow X$ is a closed immersion, then

$$Z \cap U$$

is affine and there exists an ideal

$$I \subseteq A$$

such that

$$Z \cap U \cong \mathrm{Spec}(A/I).$$

On this affine chart, the ideal sheaf is

$$\mathcal{I}_{Z/X}|_U \cong \tilde{I}.$$

This immediately implies that

$$\mathcal{I}_{Z/X}$$

is quasi-coherent.

If X is Noetherian, then every ideal $I \subseteq A$ is finitely generated. Hence

$$\tilde{I}$$

is coherent on every affine open set, and therefore

$$\mathcal{I}_{Z/X}$$

is coherent.

Thus:

$$\boxed{Z \hookrightarrow X \text{ closed immersion} \implies \mathcal{I}_{Z/X} \text{ quasi-coherent,}}$$

and, if X is Noetherian,

$$\boxed{\mathcal{I}_{Z/X} \text{ coherent.}}$$

8.9 The correspondence between ideals and closed subschemes

The affine correspondence

Let

$$X = \mathrm{Spec} A.$$

Given an ideal

$$I \subseteq A,$$

the quotient map

$$A \longrightarrow A/I$$

induces a closed immersion

$$\mathrm{Spec}(A/I) \hookrightarrow \mathrm{Spec} A.$$

Conversely, every closed immersion into $\mathrm{Spec} A$ is isomorphic to one obtained in this way.

Thus there is a bijection:

$$\boxed{\{\text{closed subschemes of } \mathrm{Spec} A\} \longleftrightarrow \{\text{ideals } I \subseteq A\} .}$$

The underlying closed subset of

$$\mathrm{Spec}(A/I)$$

is

$$V(I) = V(\sqrt{I}).$$

Thus the reduced closed subscheme associated to I is defined by $\sqrt{I}$, while the possible nilpotent thickening is contained in the difference between I and $\sqrt{I}$.

The global correspondence

On a general scheme X , one replaces ideals of a single ring by quasi-coherent ideal sheaves

$$\mathcal{I} \subseteq \mathcal{O}_X.$$

Given a closed subscheme

$$i : Z \hookrightarrow X,$$

one obtains its ideal sheaf

$$\mathcal{I}_{Z/X} = \ker(\mathcal{O}_X \rightarrow i_* \mathcal{O}_Z).$$

Conversely, suppose

$$\mathcal{I} \subseteq \mathcal{O}_X$$

is a quasi-coherent ideal sheaf.

On an affine open subset

$$U = \mathrm{Spec} A \subseteq X,$$

there exists an ideal

$$I_U \subseteq A$$

such that

$$\mathcal{I}|_U \cong \widetilde{I_U}.$$

Define locally

$$Z \cap U = \mathrm{Spec}(A/I_U).$$

Because $\mathcal{I}$ is a sheaf and its local ideals restrict compatibly, these affine closed subschemes agree on overlaps and glue to a global closed subscheme

$$Z \hookrightarrow X.$$

Thus:

$$\boxed{\{\text{closed subschemes of } X\} \longleftrightarrow \{\text{quasi-coherent ideal sheaves } \mathcal{I} \subseteq \mathcal{O}_X\} .}$$

The structure sheaf of a closed subscheme

If a closed subscheme $Z \hookrightarrow X$ is defined by an ideal sheaf

$$\mathcal{I} \subseteq \mathcal{O}_X,$$

then its structure sheaf is locally the quotient

$$\mathcal{O}_X/\mathcal{I}.$$

More precisely, one has

$$i_*\mathcal{O}_Z \cong \mathcal{O}_X/\mathcal{I}.$$

Hence the defining exact sequence is

$$\boxed{0 \longrightarrow \mathcal{I} \longrightarrow \mathcal{O}_X \longrightarrow i_*\mathcal{O}_Z \longrightarrow 0.}$$

This exact sequence is one of the basic constructions in scheme theory. It allows geometric questions about the subscheme Z to be studied by relating its structure sheaf to the ambient scheme X .

For example, infinitesimal information about how Z sits inside X is encoded by the quotient

$$\mathcal{I}/\mathcal{I}^2,$$

called the *conormal sheaf* of Z in X .

Its dual, when locally free, is the normal bundle:

$$\mathcal{N}_{Z/X} = \mathcal{H}om_{\mathcal{O}_Z}(\mathcal{I}/\mathcal{I}^2, \mathcal{O}_Z).$$

Thus ideal sheaves are not only devices for defining closed subschemes; they are also the starting point for tangent, normal, and infinitesimal geometry.

8.10 Conceptual comparison of the main ideas

The four groups of results studied above fit together as follows.

Topic	Local algebraic object	Global geometric meaning
Divisor class groups	Height-one primes and valuations in normal domains	Failure of codimension-one subvarieties to be globally principal
Morphisms $\mathbb{P}^n \rightarrow \mathbb{P}^m$	Base-point-free homogeneous polynomials of one degree	Rigidity of projective space and preservation of full dimension
Quasi-coherent sheaves	A -modules on $\text{Spec}(A)$	Sheaves that can be studied by affine algebra
Coherent sheaves	Finitely generated or finitely presented modules	Finite geometric data stable under suitable operations
Closed subschemes	Ideals $I \subseteq A$ and quotients A/I	Closed sets together with nilpotent and infinitesimal structure

The repeated pattern is:

geometric construction on schemes	$\longleftrightarrow$	algebraic construction on affine coordinate rings.
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In the divisor-class problems, removing divisors reduces a global calculation to units in a factorial ring.

In the projective-morphism problem, pulling back the hyperplane bundle reduces a morphism to homogeneous polynomials and their common zero locus.

In the sheaf problems, kernels, images, cokernels, and pullbacks reduce locally to the corresponding operations on modules.

In the closed-subscheme problem, a geometric embedding is exactly a quotient construction

$$A \longrightarrow A/I$$

performed compatibly on affine charts.

These are basic examples of the general philosophy of scheme theory: schemes are global geometric spaces built from rings, and the most effective proofs move systematically between these two perspectives.

9 Examples for Problems 4–7: Divisor Classes, Projective Morphisms, Coherent Sheaves, and Closed Subschemes

The following examples illustrate the four main themes developed in the preceding theory section:

$\mathrm{Cl}(X)$, $\mathbf{P}^n \longrightarrow \mathbf{P}^m$, quasi-coherent and coherent sheaves, closed subschemes and ideal sheaves

The examples begin with elementary affine schemes and then proceed to the projective surfaces, affine cones, morphisms, and ideal sheaves occurring directly in the problems.

9.1 Examples of divisor class groups

The affine line: $\mathrm{Cl}(\mathbf{A}_k^1) = 0$

Example.

Let

$$X = \mathbf{A}_k^1 = \mathrm{Spec} k[t].$$

Since $k[t]$ is a principal ideal domain, it is in particular a unique factorization domain. Every height-one prime ideal is generated by one irreducible polynomial. Hence every prime divisor is principal, and therefore

$$\boxed{\mathrm{Cl}(\mathbf{A}_k^1) = 0.}$$

Assume for simplicity that k is algebraically closed. Then the closed points are

$$V(t - a), \quad a \in k.$$

Each point is principal:

$$\mathrm{div}(t - a) = [V(t - a)].$$

For example, the divisor

$$2[V(t)] - 3[V(t - 1)]$$

is principal because

$$\mathrm{div} \left(\frac{t^2}{(t - 1)^3} \right) = 2[V(t)] - 3[V(t - 1)].$$

Thus every formal combination of closed points on $\mathbf{A}_k^1$ is represented by the zeros and poles of a rational function.

Remark 9.1.1. If k is not algebraically closed, the codimension-one points correspond to irreducible polynomials in $k[t]$, rather than only to linear polynomials $t - a$. The conclusion

$$\mathrm{Cl}(\mathbf{A}_k^1) = 0$$

remains unchanged because $k[t]$ is still a UFD.

The affine plane: $\text{Cl}(\mathbf{A}_k^2) = 0$

Example.

Let

$$X = \mathbf{A}_k^2 = \text{Spec } k[x, y].$$

Prime divisors on X correspond to irreducible polynomial equations

$$f(x, y) = 0.$$

Examples include

$$V(x), \quad V(y), \quad V(x - y), \quad V(x^2 + y^2 - 1),$$

provided the defining polynomial is irreducible over k .

Because

$$k[x, y]$$

is a unique factorization domain, every irreducible codimension-one closed subset is principal. Hence

$$\boxed{\text{Cl}(\mathbf{A}_k^2) = 0.}$$

For instance, the divisor

$$D = 2V(x) - V(y) - V(x - y)$$

is principal:

$$D = \text{div} \left(\frac{x^2}{y(x - y)} \right).$$

This example explains why affine spaces are useful open subsets in divisor-class computations: once a scheme is reduced to an affine factorial open part, all remaining class-group information must lie on the removed boundary.

The projective line: $\text{Cl}(\mathbf{P}_k^1) \cong \mathbf{Z}$

Example.

Let

$$X = \mathbf{P}_k^1.$$

Choose the point at infinity

$$\infty = [1 : 0].$$

Its complement is

$$\mathbf{P}_k^1 \setminus \{\infty\} \cong \mathbf{A}_k^1.$$

Since

$$\text{Cl}(\mathbf{A}_k^1) = 0,$$

the class of the point at infinity generates the class group:

$$\mathbf{Z}[\infty] \rightarrow \mathrm{Cl}(\mathbf{P}_k^1).$$

On the affine chart with coordinate

$$t = \frac{X_0}{X_1},$$

the rational function $t - a$, for $a \in k$, has a zero at the point a and a pole at infinity:

$$\mathrm{div}(t - a) = [a] - [\infty].$$

Therefore every k -rational point is linearly equivalent to ∞ :

$$[a] \sim [\infty].$$

More generally, the degree of a divisor

$$D = \sum_i n_i [p_i]$$

is

$$\deg D = \sum_i n_i [k(p_i) : k].$$

Every principal divisor has degree zero. On $\mathbf{P}_k^1$, degree classifies divisor classes, giving

$$\boxed{\mathrm{Cl}(\mathbf{P}_k^1) \cong \mathbf{Z}.}$$

For example,

$$[0] + [1] - 2[\infty]$$

is principal because

$$\mathrm{div}(t(t-1)) = [0] + [1] - 2[\infty].$$

Thus, unlike the affine line, the projective line has a nontrivial class group: the missing point at infinity supplies one free divisor class.

The surface $\mathbf{P}_k^1 \times \mathbf{P}_k^1$

Example.

Let

$$X = \mathbf{P}_k^1 \times \mathbf{P}_k^1.$$

There are two natural types of irreducible curves:

$$F_1 = \{p\} \times \mathbf{P}_k^1, \quad F_2 = \mathbf{P}_k^1 \times \{q\}.$$

These are the two families of rulings of the surface.

Choose

$$D_1 = \{\infty\} \times \mathbf{P}_k^1, \quad D_2 = \mathbf{P}_k^1 \times \{\infty\}.$$

Removing them gives

$$X \setminus (D_1 \cup D_2) \cong \mathbf{A}_k^1 \times \mathbf{A}_k^1 \cong \mathbf{A}_k^2.$$

Since

$$\mathrm{Cl}(\mathbf{A}_k^2) = 0,$$

the classes $[D_1]$ and $[D_2]$ generate $\mathrm{Cl}(X)$.

Suppose there were a relation

$$a[D_1] + b[D_2] = 0.$$

Then

$$aD_1 + bD_2 = \mathrm{div}(f)$$

for some rational function f . Since the divisor of f is supported entirely on $D_1 \cup D_2$, the restriction of f to

$$\mathbf{A}_k^2$$

must be a unit. But

$$k[s, t]^\times = k^\times.$$

Hence f must be constant, so

$$\mathrm{div}(f) = 0.$$

Therefore

$$a = b = 0.$$

Consequently,

$$\boxed{\mathrm{Cl}(\mathbf{P}_k^1 \times \mathbf{P}_k^1) \cong \mathbf{Z} \oplus \mathbf{Z}.}$$

A divisor class may be represented by a pair

$$(a, b).$$

The class

$$(1, 0)$$

is represented by a divisor of the form

$$\{p\} \times \mathbf{P}_k^1,$$

while the class

$$(0, 1)$$

is represented by a divisor of the form

$$\mathbf{P}_k^1 \times \{q\}.$$

For example, the diagonal

$$\Delta = \{([S_0 : S_1], [T_0 : T_1]) : S_0T_1 - S_1T_0 = 0\}$$

is defined by an equation of bidegree $(1, 1)$. Therefore

$$[\Delta] = (1, 1).$$

Likewise, a bihomogeneous equation of bidegree $(2, 3)$ defines a divisor whose class is

$$(2, 3).$$

Principal divisors on $\mathbf{P}_k^1 \times \mathbf{P}_k^1$ **Example.**

Choose affine coordinates

$$s = \frac{S_0}{S_1}, \quad t = \frac{T_0}{T_1}$$

on the dense affine open subset

$$\mathbf{A}_k^1 \times \mathbf{A}_k^1 \subseteq \mathbf{P}_k^1 \times \mathbf{P}_k^1.$$

Regarded as a rational function on the complete surface, s vanishes along

$$\{0\} \times \mathbf{P}_k^1$$

and has a pole along

$$\{\infty\} \times \mathbf{P}_k^1.$$

Hence

$$\operatorname{div}(s) = (\{0\} \times \mathbf{P}_k^1) - (\{\infty\} \times \mathbf{P}_k^1).$$

Thus all vertical rulings are linearly equivalent.

Similarly,

$$\operatorname{div}(t) = (\mathbf{P}_k^1 \times \{0\}) - (\mathbf{P}_k^1 \times \{\infty\}),$$

so all horizontal rulings are linearly equivalent.

However, no rational function gives a relation identifying a vertical ruling with a horizontal ruling. Thus the two directions remain independent in the class group.

The quadric cone $xy = z^2$ **Example.**

Let

$$X = \operatorname{Spec} A, \quad A = k[x, y, z]/(xy - z^2).$$

Consider the prime divisor

$$D = V(x, z).$$

Its coordinate ring is

$$A/(x, z) \cong k[y],$$

so

$$D \cong \mathbf{A}_k^1.$$

Now consider the open subset

$$U = D(x) \subseteq X.$$

Since x is invertible on U , the defining equation gives

$$y = \frac{z^2}{x}.$$

Therefore

$$\Gamma(U, \mathcal{O}_U) \cong k[x, x^{-1}, z].$$

This is a factorial ring, so

$$\text{Cl}(U) = 0.$$

The complement $V(x)$ has only one irreducible codimension-one component, namely D . Thus $[D]$ generates $\text{Cl}(X)$.

At the generic point of D , the element y is invertible, and

$$x = \frac{z^2}{y}.$$

Hence x vanishes twice along D :

$$\text{ord}_D(x) = 2.$$

Therefore

$$\text{div}(x) = 2D.$$

Consequently,

$$2[D] = 0.$$

The divisor D itself is not principal, so its class is nonzero. Therefore

$$\boxed{\text{Cl}(X) \cong \mathbf{Z}/2\mathbf{Z}.$$

There is another ruling line

$$D' = V(y, z).$$

Since

$$\text{div}(z) = D + D',$$

we obtain

$$[D'] = -[D].$$

But in a group of order two,

$$-[D] = [D].$$

Therefore both ruling lines represent the same nontrivial divisor class:

$$[D'] = [D].$$

The family $xy = z^n$

Example.

The preceding example extends naturally to the family

$$X_n = \text{Spec } k[x, y, z]/(xy - z^n), \quad n \geq 2.$$

Let

$$D = V(x, z).$$

On the open subset

$$D(x) \subseteq X_n,$$

one has

$$y = \frac{z^n}{x},$$

and therefore

$$\Gamma(D(x), \mathcal{O}_{X_n}) \cong k[x, x^{-1}, z].$$

Thus the complement is factorial, and $[D]$ generates the class group.

At the generic point of D , the element y is a unit, and

$$x = \frac{z^n}{y}.$$

Therefore

$$\operatorname{div}(x) = nD.$$

Hence

$$n[D] = 0.$$

The same unit calculation as before shows that this is the only relation. Consequently,

$$\boxed{\operatorname{Cl}(X_n) \cong \mathbf{Z}/n\mathbf{Z}.$$

In particular,

$$xy = z^2 \implies \operatorname{Cl}(X_2) \cong \mathbf{Z}/2\mathbf{Z},$$

$$xy = z^3 \implies \operatorname{Cl}(X_3) \cong \mathbf{Z}/3\mathbf{Z},$$

and

$$xy = z^4 \implies \operatorname{Cl}(X_4) \cong \mathbf{Z}/4\mathbf{Z}.$$

This family gives a useful collection of normal affine surface singularities with finite cyclic divisor class groups.

The three-dimensional quadric cone $xy = zw$

Example.

Let

$$X = \operatorname{Spec} A, \quad A = k[x, y, z, w]/(xy - zw).$$

On the open subset

$$U = D(x),$$

one may solve for y :

$$y = \frac{zw}{x}.$$

Hence

$$\Gamma(U, \mathcal{O}_U) \cong k[x, x^{-1}, z, w].$$

This is factorial, so

$$\mathrm{Cl}(U) = 0.$$

The complement $V(x)$ has two irreducible codimension-one components:

$$D = V(x, z), \quad E = V(x, w).$$

Indeed,

$$A/(x) \cong k[y, z, w]/(zw),$$

whose minimal primes are (z) and (w) .

Their coordinate rings are

$$A/(x, z) \cong k[y, w],$$

and

$$A/(x, w) \cong k[y, z].$$

Therefore

$$D \cong \mathbf{A}_k^2, \quad E \cong \mathbf{A}_k^2.$$

At the generic point of D , the functions y and w are invertible, and

$$x = \frac{w}{y}z.$$

Hence

$$\mathrm{ord}_D(x) = 1.$$

Similarly, at the generic point of E ,

$$x = \frac{z}{y}w,$$

so

$$\mathrm{ord}_E(x) = 1.$$

Therefore

$$\mathrm{div}(x) = D + E.$$

Consequently,

$$[D] + [E] = 0.$$

This is the only relation between the two generators, and hence

$$\boxed{\mathrm{Cl}(X) \cong \mathbf{Z}.}$$

One may take $[D]$ as a generator, with

$$[E] = -[D].$$

Comparison of the class-group examples

The examples above may be summarized as follows:

Scheme	Factorial open sub-set	Principal relation	Class group
$\mathbf{A}_k^1$	$\mathbf{A}_k^1$	Every divisor is principal	$\mathbf{0}$
$\mathbf{A}_k^2$	$\mathbf{A}_k^2$	Every divisor is principal	$\mathbf{0}$
$\mathbf{P}_k^1$	$\mathbf{A}_k^1$	$[a] - [\infty] = \text{div}(t - a)$	$\mathbf{Z}$
$\mathbf{P}_k^1 \times \mathbf{P}_k^1$	$\mathbf{A}_k^2$	No relation between the two rulings	$\mathbf{Z}^2$
$\text{Spec } k[x, y, z]/(xy - z^2)$	$\text{Spec } k[x, x^{-1}, z]$	$\text{div}(x) = 2D$	$\mathbf{Z}/2\mathbf{Z}$
$\text{Spec } k[x, y, z]/(xy - z^n)$	$\text{Spec } k[x, x^{-1}, z]$	$\text{div}(x) = nD$	$\mathbf{Z}/n\mathbf{Z}$
$\text{Spec } k[x, y, z, w]/(xy - zw)$	$\text{Spec } k[x, x^{-1}, z, w]$	$\text{div}(x) = D + E$	$\mathbf{Z}$

The contrast between the two cone examples is especially important:

$$\text{div}(x) = 2D$$

produces a torsion class, while

$$\text{div}(x) = D + E$$

produces one relation between two free generators.

9.2 Examples of morphisms between projective spaces

A constant morphism

Example.

Define

$$\varphi : \mathbf{P}_k^2 \longrightarrow \mathbf{P}_k^3$$

by

$$\varphi([X : Y : Z]) = [1 : 0 : 0 : 0].$$

Every point is sent to the same image, so

$$\varphi(\mathbf{P}_k^2) = \{[1 : 0 : 0 : 0]\}.$$

Thus

$$\dim \varphi(\mathbf{P}_k^2) = 0.$$

The pullback of the hyperplane line bundle is trivial:

$$\varphi^* \mathcal{O}_{\mathbf{P}^3}(1) \cong \mathcal{O}_{\mathbf{P}^2}.$$

This is the degree-zero case in the general theory of morphisms from projective space.

A linear embedding $\mathbf{P}_k^2 \hookrightarrow \mathbf{P}_k^3$ **Example.**

Define

$$\varphi : \mathbf{P}_k^2 \longrightarrow \mathbf{P}_k^3$$

by

$$\varphi([X : Y : Z]) = [X : Y : Z : 0].$$

The image is the projective plane

$$V(W) \subseteq \mathbf{P}_k^3.$$

Therefore

$$\varphi(\mathbf{P}_k^2) \cong \mathbf{P}_k^2,$$

and

$$\dim \varphi(\mathbf{P}_k^2) = 2.$$

The morphism is given by linear forms, and therefore

$$\varphi^* \mathcal{O}_{\mathbf{P}^3}(1) \cong \mathcal{O}_{\mathbf{P}^2}(1).$$

This is a basic example of a nonconstant morphism whose image has the full dimension of the source.

A degree- d morphism $\mathbf{P}_k^1 \rightarrow \mathbf{P}_k^1$ **Example.**

For an integer $d \geq 1$, define

$$\varphi_d : \mathbf{P}_k^1 \longrightarrow \mathbf{P}_k^1$$

by

$$\varphi_d([S : T]) = [S^d : T^d].$$

The two homogeneous polynomials S^d and T^d never vanish simultaneously on $\mathbf{P}_k^1$. Hence this is a well-defined morphism.

On the affine chart $T \neq 0$, with coordinate

$$u = \frac{S}{T},$$

the map is

$$u \longmapsto u^d.$$

The pullback of the hyperplane bundle is

$$\varphi_d^* \mathcal{O}_{\mathbf{P}^1}(1) \cong \mathcal{O}_{\mathbf{P}^1}(d).$$

The image is all of $\mathbf{P}_k^1$, so

$$\dim \varphi_d(\mathbf{P}_k^1) = 1.$$

For $d = 2$, the map

$$[S : T] \mapsto [S^2 : T^2]$$

is a degree-two morphism. It may identify different points, but it does not lower the dimension of the image.

The Veronese embedding of $\mathbf{P}_k^1$ as a conic

Example.

Define

$$\nu_2 : \mathbf{P}_k^1 \longrightarrow \mathbf{P}_k^2$$

by

$$\nu_2([S : T]) = [S^2 : ST : T^2].$$

The three homogeneous coordinates never vanish simultaneously, so this defines a morphism everywhere on $\mathbf{P}_k^1$.

Write the homogeneous coordinates on $\mathbf{P}_k^2$ as

$$[X : Y : Z].$$

For every point in the image,

$$XZ - Y^2 = (S^2)(T^2) - (ST)^2 = 0.$$

Thus

$$\nu_2(\mathbf{P}_k^1) \subseteq V(XZ - Y^2).$$

Conversely, the smooth conic

$$C = V(XZ - Y^2) \subseteq \mathbf{P}_k^2$$

is parametrized by this map. Therefore

$$\nu_2(\mathbf{P}_k^1) = C.$$

The image is one-dimensional:

$$\dim C = 1.$$

Moreover,

$$\nu_2^* \mathcal{O}_{\mathbf{P}^2}(1) \cong \mathcal{O}_{\mathbf{P}^1}(2).$$

This example shows that a nonconstant projective morphism may transform a projective line into a curved-looking embedded conic while preserving its dimension.

The Veronese embedding of $\mathbf{P}_k^2$

Example.

Define

$$\nu_2 : \mathbf{P}_k^2 \longrightarrow \mathbf{P}_k^5$$

by

$$\nu_2([X : Y : Z]) = [X^2 : XY : XZ : Y^2 : YZ : Z^2].$$

The six coordinates are homogeneous of degree 2, and they have no common zero in $\mathbf{P}_k^5$. Hence ν_2 is a morphism.

Its image is a projective surface in $\mathbf{P}_k^5$, called the Veronese surface. Since ν_2 is an embedding,

$$\dim \nu_2(\mathbf{P}_k^2) = 2.$$

The associated pullback relation is

$$\nu_2^* \mathcal{O}_{\mathbf{P}^5}(1) \cong \mathcal{O}_{\mathbf{P}^2}(2).$$

Again, the image of a nonconstant morphism from projective space has the full dimension of the source.

A finite morphism $\mathbf{P}_k^2 \rightarrow \mathbf{P}_k^2$

Example.

For $d \geq 1$, define

$$\varphi_d : \mathbf{P}_k^2 \longrightarrow \mathbf{P}_k^2$$

by

$$\varphi_d([X : Y : Z]) = [X^d : Y^d : Z^d].$$

The homogeneous coordinates X^d, Y^d, Z^d have no common projective zero, so this is a morphism.

Its image is all of $\mathbf{P}_k^2$, and therefore

$$\dim \varphi_d(\mathbf{P}_k^2) = 2.$$

For $d > 1$, this morphism generally fails to be injective. Nevertheless, its image still has full dimension.

Thus the conclusion of the projective-morphism theorem is not that every nonconstant map is an embedding. Rather, it says that a nonconstant morphism from $\mathbf{P}_k^n$ cannot have a positive-dimensional image of dimension strictly less than n .

Why $[X : Y]$ does not define a morphism $\mathbf{P}_k^2 \rightarrow \mathbf{P}_k^1$

Example.

One might try to define a map

$$\psi : \mathbf{P}_k^2 \dashrightarrow \mathbf{P}_k^1$$

by

$$\psi([X : Y : Z]) = [X : Y].$$

This expression is valid whenever

$$(X, Y) \neq (0, 0).$$

However, at the point

$$[0 : 0 : 1] \in \mathbf{P}_k^2,$$

both proposed target coordinates vanish:

$$[X : Y] = [0 : 0].$$

But $[0 : 0]$ is not a point of projective space. Therefore the formula is not defined at $[0 : 0 : 1]$.

Thus

$$[X : Y : Z] \mapsto [X : Y]$$

defines only a rational map

$$\mathbf{P}_k^2 \dashrightarrow \mathbf{P}_k^1,$$

not a morphism on all of $\mathbf{P}_k^2$.

This gives a concrete explanation of the general theorem:

there is no nonconstant everywhere-defined morphism $\mathbf{P}_k^2 \longrightarrow \mathbf{P}_k^1$.

The point $[0 : 0 : 1]$ is called a base point of the proposed linear system.

Higher-degree coordinates do not automatically remove base points

Example.

Consider the formula

$$[X : Y : Z] \mapsto [X^2 : XY : Y^2].$$

All three target coordinates have the same degree. However, at the point

$$[0 : 0 : 1],$$

they all vanish:

$$X^2 = XY = Y^2 = 0.$$

Therefore this formula again gives only a rational map

$$\mathbf{P}_k^2 \dashrightarrow \mathbf{P}_k^2,$$

not a morphism defined everywhere.

This illustrates the essential requirement for a projective morphism:

homogeneous coordinates of one degree must have no common zero.

9.3 Examples of quasi-coherent and coherent sheaves

The structure sheaf of an affine scheme

Example.

Let

$$X = \operatorname{Spec} A.$$

The structure sheaf is the sheaf associated to the A -module A :

$$\mathcal{O}_X \cong \tilde{A}.$$

Therefore $\mathcal{O}_X$ is quasi-coherent.

If A is Noetherian, then A is finitely generated as a module over itself. Hence

$$\mathcal{O}_X$$

is coherent.

For example,

$$\mathcal{O}_{\mathbf{A}_k^2}$$

is coherent because

$$\mathbf{A}_k^2 = \operatorname{Spec} k[x, y]$$

and $k[x, y]$ is Noetherian.

The ideal sheaf of the origin in $\mathbf{A}_k^2$

Example.

Let

$$X = \mathbf{A}_k^2 = \operatorname{Spec} k[x, y].$$

The origin is defined by the maximal ideal

$$I = (x, y).$$

The associated ideal sheaf is

$$\widetilde{(x, y)} \subseteq \mathcal{O}_{\mathbf{A}_k^2}.$$

The quotient is

$$k[x, y]/(x, y) \cong k,$$

which is the coordinate ring of the reduced origin.

Thus there is an exact sequence

$$0 \longrightarrow \widetilde{(x, y)} \longrightarrow \mathcal{O}_{\mathbf{A}_k^2} \longrightarrow \mathcal{O}_{\{(0,0)\}} \longrightarrow 0.$$

Since the ideal (x, y) is finitely generated, its associated sheaf is coherent.

This example shows concretely how a closed point is described by a coherent ideal sheaf.

Multiplication by t on the affine line

Example.

Let

$$X = \mathbf{A}_k^1 = \operatorname{Spec} k[t].$$

Consider the morphism

$$\mathcal{O}_X \xrightarrow{\cdot t} \mathcal{O}_X.$$

On modules, this is the map

$$k[t] \xrightarrow{\cdot t} k[t].$$

Since $k[t]$ is a domain,

$$\ker(\cdot t) = 0.$$

Its image is the ideal

$$(t) \subseteq k[t].$$

Its cokernel is

$$k[t]/(t) \cong k.$$

Therefore, as sheaves,

$$\ker(\cdot t) = 0,$$

$$\operatorname{im}(\cdot t) = \widetilde{(t)},$$

and

$$\operatorname{coker}(\cdot t) \cong \mathcal{O}_{\{0\}}.$$

Hence one obtains the exact sequence

$$0 \longrightarrow \mathcal{O}_{\mathbf{A}_k^1} \xrightarrow{\cdot t} \mathcal{O}_{\mathbf{A}_k^1} \longrightarrow \mathcal{O}_{\{0\}} \longrightarrow 0.$$

This sequence says that when regular functions are taken modulo those divisible by t , only their value at the origin remains.

The ideal of the origin as an image

Example.

Let

$$X = \mathbf{A}_k^2 = \operatorname{Spec} k[x, y].$$

Define a morphism

$$\varphi : \mathcal{O}_X^{\oplus 2} \longrightarrow \mathcal{O}_X$$

by

$$\varphi(a, b) = ax + by.$$

On global modules, this is

$$k[x, y]^2 \longrightarrow k[x, y], \quad (a, b) \longmapsto ax + by.$$

Its image is precisely the ideal

$$(x, y).$$

Thus

$$\operatorname{im} \varphi = \widetilde{(x, y)}.$$

The cokernel is

$$k[x, y]/(x, y) \cong k,$$

and consequently

$$\operatorname{coker} \varphi \cong \mathcal{O}_{\{(0,0)\}}.$$

The kernel is generated by the relation

$$(-y, x),$$

since

$$(-y)x + xy = 0.$$

Therefore there is an exact sequence

$$0 \longrightarrow \mathcal{O}_{\mathbf{A}_k^2} \xrightarrow{\begin{pmatrix} -y \\ x \end{pmatrix}} \mathcal{O}_{\mathbf{A}_k^2}^{\oplus 2} \xrightarrow{\begin{pmatrix} x & y \end{pmatrix}} \mathcal{O}_{\mathbf{A}_k^2} \longrightarrow \mathcal{O}_{\{(0,0)\}} \longrightarrow 0.$$

This is the Koszul resolution of the structure sheaf of the origin in $\mathbf{A}_k^2$.

A quasi-coherent sheaf that is not coherent

Example.

Let

$$X = \operatorname{Spec} k.$$

A quasi-coherent sheaf on X is simply a k -vector space, viewed as a sheaf on the one-point scheme.

Consider the k -vector space

$$M = k[t].$$

The associated sheaf

$$\widetilde{k[t]}$$

is quasi-coherent.

However,

$$k[t]$$

is infinite-dimensional as a vector space over k . Hence it is not finitely generated as a k -module.

Therefore

$\widetilde{k[t]}$ is quasi-coherent but not coherent on $\operatorname{Spec} k$.

This is exactly the sheaf that occurs as the pushforward

$$p_* \mathcal{O}_{\mathbf{A}_k^1}$$

under the structure morphism

$$p : \mathbf{A}_k^1 \longrightarrow \operatorname{Spec} k.$$

A coherent skyscraper sheaf

Example.

Let

$$X = \mathbf{A}_k^1 = \operatorname{Spec} k[t].$$

The origin is defined by the ideal

$$(t).$$

Its structure sheaf, regarded as a sheaf on X , is

$$\mathcal{O}_{\{0\}} \cong \widetilde{k[t]/(t)}.$$

Since

$$k[t]/(t) \cong k$$

is finitely generated as a $k[t]$ -module, the sheaf

$$\mathcal{O}_{\{0\}}$$

is coherent.

Likewise,

$$\widetilde{k[t]/(t^2)}$$

is coherent, but it describes a nonreduced thickened point rather than the ordinary reduced point.

Thus coherence does not require a scheme to be reduced. It expresses finite generation, not the absence of nilpotents.

9.4 Examples of pullbacks and pushforwards

Pulling back the origin under the squaring map

Example.

Let

$$g : \mathbf{A}_k^1 \longrightarrow \mathbf{A}_k^1$$

be defined by

$$u \longmapsto u^2.$$

Writing the coordinate on the target as t , this morphism corresponds to the ring homomorphism

$$k[t] \longrightarrow k[u], \quad t \longmapsto u^2.$$

Let

$$Z = V(t) \subseteq \mathbf{A}_k^1$$

be the origin in the target. Its structure sheaf, as an $\mathcal{O}_{\mathbf{A}_k^1}$ -module, is

$$\mathcal{O}_Z \cong \widetilde{k[t]/(t)}.$$

Its pullback is

$$g^* \mathcal{O}_Z \cong \left(\widetilde{k[t]/(t)} \right) \otimes_{k[t]} k[u].$$

Since t acts on $k[u]$ as multiplication by u^2 , one obtains

$$\left(\widetilde{k[t]/(t)} \right) \otimes_{k[t]} k[u] \cong k[u]/(u^2).$$

Therefore

$$\boxed{g^* \mathcal{O}_Z \cong \widetilde{k[u]/(u^2)}}.$$

Set-theoretically, the inverse image of 0 is only the point $u = 0$. Scheme-theoretically, however, the inverse image is the double point

$$\mathrm{Spec} \, k[u]/(u^2).$$

Thus the pullback remembers the multiplicity of the zero of the function u^2 .

Pulling back the origin under a cubic map

Example.

Let

$$g : \mathbf{A}_k^1 \longrightarrow \mathbf{A}_k^1$$

be defined by

$$t \longmapsto u^3.$$

Equivalently, on coordinate rings,

$$k[t] \longrightarrow k[u], \quad t \longmapsto u^3.$$

Pulling back the origin $V(t)$ gives

$$k[u]/(u^3).$$

Thus the scheme-theoretic inverse image is

$$\mathrm{Spec} \, k[u]/(u^3),$$

a triple point at the origin.

The underlying topological inverse image is still just

$$\{0\},$$

but its scheme structure records multiplicity three.

Pullback of $\mathcal{O}_{\mathbf{P}^1}(1)$ under a degree- d map**Example.**

Let

$$\varphi_d : \mathbf{P}_k^1 \longrightarrow \mathbf{P}_k^1$$

be defined by

$$\varphi_d([S : T]) = [S^d : T^d].$$

Then

$$\varphi_d^* \mathcal{O}_{\mathbf{P}^1}(1) \cong \mathcal{O}_{\mathbf{P}^1}(d).$$

For instance, if

$$d = 2,$$

then

$$\varphi_2^* \mathcal{O}_{\mathbf{P}^1}(1) \cong \mathcal{O}_{\mathbf{P}^1}(2).$$

This may also be seen directly from divisors. The point

$$[0 : 1] \in \mathbf{P}_k^1$$

is defined by the equation

$$S = 0.$$

Under the degree-two map, its defining equation pulls back to

$$S^2 = 0.$$

Thus the pullback divisor has multiplicity two:

$$\varphi_2^*([0 : 1]) = 2[0 : 1].$$

More generally,

$$\varphi_d^*([0 : 1]) = d[0 : 1].$$

A pushforward that is not coherent**Example.**

Consider the structure morphism

$$p : \mathbf{A}_k^1 \longrightarrow \operatorname{Spec} k.$$

The sheaf

$$\mathcal{O}_{\mathbf{A}_k^1}$$

is coherent.

However,

$$p_* \mathcal{O}_{\mathbf{A}_k^1}$$

corresponds to the k -vector space

$$\Gamma(\mathbf{A}_k^1, \mathcal{O}_{\mathbf{A}_k^1}) = k[t].$$

As a vector space over k ,

$$k[t] = k \oplus kt \oplus kt^2 \oplus kt^3 \oplus \cdots$$

is infinite-dimensional.

A coherent sheaf on $\operatorname{Spec} k$ corresponds to a finite-dimensional k -vector space. Therefore

$$\boxed{p_* \mathcal{O}_{\mathbf{A}_k^1} \text{ is not coherent.}}$$

It remains quasi-coherent, because it is still the sheaf associated to the k -module $k[t]$.

A positive pushforward example: a finite morphism

Example.

Let

$$g : \mathbf{A}_k^1 \longrightarrow \mathbf{A}_k^1$$

be defined by

$$u \longmapsto u^2.$$

Write the coordinate on the target as t . The corresponding ring map is

$$k[t] \longrightarrow k[u], \quad t \longmapsto u^2.$$

As a module over $k[t]$, the ring $k[u]$ is generated by

$$1 \quad \text{and} \quad u.$$

Indeed, every polynomial $F(u) \in k[u]$ can be written uniquely in the form

$$F(u) = A(u^2) + uB(u^2)$$

for some polynomials $A, B \in k[t]$. Hence

$$k[u] \cong k[t] \oplus u k[t]$$

as a $k[t]$ -module.

Thus $k[u]$ is finite over $k[t]$, and therefore

$$g_* \mathcal{O}_{\mathbf{A}_k^1}$$

is coherent.

This should be contrasted with

$$\mathbf{A}_k^1 \longrightarrow \operatorname{Spec} k,$$

where the pushforward of the structure sheaf is not coherent.

9.5 Examples of closed subschemes and ideal sheaves

A reduced point in $\mathbf{A}_k^1$

Example.

Let

$$X = \mathbf{A}_k^1 = \operatorname{Spec} k[x].$$

The ideal

$$I = (x)$$

defines the closed subscheme

$$Z = \operatorname{Spec} k[x]/(x).$$

Since

$$k[x]/(x) \cong k,$$

we have

$$Z \cong \operatorname{Spec} k.$$

This is the ordinary reduced point at the origin.

Its ideal sheaf is

$$\mathcal{I}_{Z/X} = \widetilde{(x)}.$$

The defining exact sequence is

$$0 \longrightarrow \widetilde{(x)} \longrightarrow \mathcal{O}_{\mathbf{A}_k^1} \longrightarrow \mathcal{O}_{\{0\}} \longrightarrow 0.$$

A double point in $\mathbf{A}_k^1$

Example.

Now take the ideal

$$I = (x^2).$$

It defines the closed subscheme

$$Z = \operatorname{Spec} k[x]/(x^2).$$

Its underlying topological space is still the single point 0, because

$$V(x^2) = V(x).$$

However, its coordinate ring contains a nonzero nilpotent element:

$$x \neq 0, \quad x^2 = 0 \quad \text{in } k[x]/(x^2).$$

Therefore Z is not reduced. It is a double point, or first infinitesimal thickening of the origin.

The reduced point and the double point have the same underlying topological subset but different scheme structures:

$$\operatorname{Spec} k[x]/(x) \not\cong \operatorname{Spec} k[x]/(x^2).$$

A triple point in $\mathbf{A}_k^1$ **Example.**

The ideal

$$I = (x^3)$$

defines

$$Z = \operatorname{Spec} k[x]/(x^3).$$

As a k -vector space,

$$k[x]/(x^3) = k \oplus kx \oplus kx^2.$$

In this ring,

$$x^3 = 0, \quad x^2 \neq 0.$$

Thus Z is a triple infinitesimal thickening of the origin.

There is a chain of closed subschemes:

$$\operatorname{Spec} k[x]/(x) \subseteq \operatorname{Spec} k[x]/(x^2) \subseteq \operatorname{Spec} k[x]/(x^3) \subseteq \mathbf{A}_k^1.$$

Each has the same underlying point, but each successive scheme remembers more nilpotent information.

Two reduced points in $\mathbf{A}_k^1$ **Example.**

Let

$$I = (x(x-1)) \subseteq k[x].$$

Then

$$Z = \operatorname{Spec} k[x]/(x(x-1)).$$

Since the ideals

$$(x) \quad \text{and} \quad (x-1)$$

are comaximal, the Chinese remainder theorem gives

$$k[x]/(x(x-1)) \cong k[x]/(x) \times k[x]/(x-1) \cong k \times k.$$

Therefore Z is the disjoint union of two reduced points:

$$Z = \{0, 1\}.$$

This is very different from

$$\operatorname{Spec} k[x]/(x^2),$$

which has only one topological point but possesses nilpotent structure.

A line in $\mathbf{A}_k^2$ **Example.**

Let

$$X = \mathbf{A}_k^2 = \operatorname{Spec} k[x, y].$$

The ideal

$$I = (y)$$

defines the x -axis:

$$Z = V(y).$$

Its coordinate ring is

$$k[x, y]/(y) \cong k[x].$$

Hence

$$Z \cong \mathbf{A}_k^1.$$

Its ideal sheaf is

$$\mathcal{I}_{Z/X} = \widetilde{(y)}.$$

The corresponding exact sequence is

$$0 \longrightarrow \widetilde{(y)} \longrightarrow \mathcal{O}_{\mathbf{A}_k^2} \longrightarrow \mathcal{O}_{V(y)} \longrightarrow 0.$$

A doubled line in $\mathbf{A}_k^2$ **Example.**

Replace the ideal (y) by

$$I = (y^2).$$

The corresponding closed subscheme is

$$Z = V(y^2).$$

Its underlying topological set is still the x -axis:

$$V(y^2) = V(y).$$

However, its coordinate ring is

$$k[x, y]/(y^2),$$

in which the class of y is nonzero but nilpotent:

$$y \neq 0, \quad y^2 = 0.$$

Thus Z is a doubled line. It contains first-order infinitesimal information in the direction normal to the ordinary line $V(y)$.

The union of the coordinate axes

Example.

In

$$\mathbf{A}_k^2 = \operatorname{Spec} k[x, y],$$

consider the ideal

$$I = (xy).$$

The corresponding closed subscheme is

$$Z = V(xy) = V(x) \cup V(y).$$

Thus Z is the union of the two coordinate axes.

Its coordinate ring is

$$k[x, y]/(xy).$$

The scheme is reduced, because the ideal (xy) is radical in the polynomial ring over a field:

$$(xy) = (x) \cap (y).$$

However, it is reducible, since it is the union of the two irreducible components

$$V(x) \quad \text{and} \quad V(y).$$

It is also singular at the origin, where the two components meet.

This example distinguishes three notions:

reduced does not imply irreducible, and irreducible does not imply smooth.

A fat point in $\mathbf{A}_k^2$

Example.

Let

$$X = \mathbf{A}_k^2 = \operatorname{Spec} k[x, y].$$

The ordinary origin is defined by

$$(x, y).$$

Now consider the square of this ideal:

$$I = (x, y)^2 = (x^2, xy, y^2).$$

The resulting closed subscheme is

$$Z = \operatorname{Spec} k[x, y]/(x^2, xy, y^2).$$

As a k -vector space, its coordinate ring has basis

$$1, \quad x, \quad y.$$

The classes of x and y are nonzero, but every second-order expression vanishes:

$$x^2 = xy = y^2 = 0.$$

Thus Z is a fat point supported at the origin. It remembers first-order infinitesimal directions in both coordinate directions.

The reduced origin

$$\operatorname{Spec} k[x, y]/(x, y)$$

remembers only the point itself, whereas

$$\operatorname{Spec} k[x, y]/(x, y)^2$$

also remembers the first-order tangent information around that point.

The diagonal in $\mathbf{A}_k^1 \times \mathbf{A}_k^1$

Example.

Let

$$X = \mathbf{A}_k^1 \times \mathbf{A}_k^1 = \operatorname{Spec} k[x, y].$$

The diagonal consists of the points satisfying

$$x = y.$$

It is therefore defined by the ideal

$$I = (x - y).$$

Its coordinate ring is

$$k[x, y]/(x - y) \cong k[x].$$

Hence the diagonal is isomorphic to the affine line:

$$\Delta \cong \mathbf{A}_k^1.$$

Its ideal sheaf is

$$\mathcal{I}_{\Delta/X} = \widetilde{(x - y)}.$$

The diagonal is a fundamental example in scheme theory. For a general scheme X , the diagonal morphism

$$\Delta_X : X \longrightarrow X \times X$$

plays a central role in the definition of separated morphisms and in the construction of differentials.

The graph of a polynomial map

Example.

Consider the morphism

$$f : \mathbf{A}_k^1 \longrightarrow \mathbf{A}_k^1, \quad x \longmapsto x^2.$$

Its graph is the subset of

$$\mathbf{A}_k^1 \times \mathbf{A}_k^1 = \operatorname{Spec} k[x, y]$$

defined by

$$y = x^2.$$

Thus the graph is the closed subscheme

$$\Gamma_f = V(y - x^2),$$

defined by the ideal

$$I = (y - x^2).$$

Its coordinate ring is

$$k[x, y]/(y - x^2) \cong k[x].$$

Therefore

$$\Gamma_f \cong \mathbf{A}_k^1.$$

This example shows that morphisms can often be studied geometrically by studying their graphs as closed subschemes of products.

A hyperplane in projective space

Example.

Let

$$H = V(X_0) \subseteq \mathbf{P}_k^n.$$

Then

$$H \cong \mathbf{P}_k^{n-1}.$$

The ideal sheaf of H is

$$\mathcal{I}_H \cong \mathcal{O}_{\mathbf{P}^n}(-1).$$

Indeed, multiplication by the defining homogeneous linear form X_0 gives the exact sequence

$$0 \longrightarrow \mathcal{O}_{\mathbf{P}^n}(-1) \xrightarrow{\cdot X_0} \mathcal{O}_{\mathbf{P}^n} \longrightarrow \mathcal{O}_H \longrightarrow 0.$$

More generally, if a hypersurface

$$Y = V(F) \subseteq \mathbf{P}_k^n$$

is defined by a homogeneous polynomial F of degree d , then

$$\mathcal{I}_Y \cong \mathcal{O}_{\mathbf{P}^n}(-d),$$

and there is an exact sequence

$$0 \longrightarrow \mathcal{O}_{\mathbf{P}^n}(-d) \xrightarrow{\cdot F} \mathcal{O}_{\mathbf{P}^n} \longrightarrow \mathcal{O}_Y \longrightarrow 0.$$

For example, the plane conic

$$C = V(XZ - Y^2) \subseteq \mathbf{P}_k^2$$

has defining equation of degree 2, so

$$\mathcal{I}_C \cong \mathcal{O}_{\mathbf{P}^2}(-2),$$

and

$$0 \longrightarrow \mathcal{O}_{\mathbf{P}^2}(-2) \xrightarrow{\cdot(XZ-Y^2)} \mathcal{O}_{\mathbf{P}^2} \longrightarrow \mathcal{O}_C \longrightarrow 0.$$

9.6 Examples connecting the different topics

The divisor $D = V(x, z)$ on the cone $xy = z^2$

Example.

Let

$$X = \operatorname{Spec} k[x, y, z]/(xy - z^2).$$

The closed subscheme

$$D = V(x, z)$$

is defined by the ideal

$$I_D = (x, z).$$

Its coordinate ring is

$$\frac{k[x, y, z]/(xy - z^2)}{(x, z)} \cong k[y].$$

Hence

$$D \cong \mathbf{A}_k^1.$$

Thus D is simultaneously:

- a closed subscheme of X , defined by the ideal sheaf associated to (x, z) ;
- an irreducible codimension-one subvariety of X , hence a prime divisor;
- a representative of the nontrivial class in

$$\operatorname{Cl}(X) \cong \mathbf{Z}/2\mathbf{Z}.$$

The relation

$$\operatorname{div}(x) = 2D$$

says that D is not itself principal, but twice D is principal.

This example joins together closed subschemes, prime divisors, rational functions, and divisor classes.

The two divisors on $xy = zw$ **Example.**

Let

$$X = \operatorname{Spec} k[x, y, z, w]/(xy - zw).$$

Consider the two closed subschemes

$$D = V(x, z), \quad E = V(x, w).$$

They are defined by the ideals

$$I_D = (x, z), \quad I_E = (x, w).$$

Their coordinate rings are

$$\frac{k[x, y, z, w]/(xy - zw)}{(x, z)} \cong k[y, w],$$

and

$$\frac{k[x, y, z, w]/(xy - zw)}{(x, w)} \cong k[y, z].$$

Therefore

$$D \cong \mathbf{A}_k^2, \quad E \cong \mathbf{A}_k^2.$$

Moreover, the zero locus of x decomposes as

$$V(x) = D \cup E.$$

At the divisor level,

$$\operatorname{div}(x) = D + E.$$

Therefore, in the divisor class group,

$$[D] + [E] = 0.$$

Hence

$$[E] = -[D].$$

This gives a direct connection between:

- decomposition of a closed subscheme into irreducible components;
- principal divisors of regular functions;
- relations inside the divisor class group.

The conic as both an image and a closed subscheme

Example.

Consider the Veronese morphism

$$\nu_2 : \mathbf{P}_k^1 \longrightarrow \mathbf{P}_k^2, \quad [S : T] \longmapsto [S^2 : ST : T^2].$$

Its image is the conic

$$C = V(XZ - Y^2) \subseteq \mathbf{P}_k^2.$$

Thus C can be described in two complementary ways:

1. As the image of a morphism from projective space:

$$C = \nu_2(\mathbf{P}_k^1).$$

2. As the closed subscheme of $\mathbf{P}_k^2$ defined by the homogeneous ideal

$$(XZ - Y^2).$$

Its ideal sheaf satisfies

$$\mathcal{I}_C \cong \mathcal{O}_{\mathbf{P}^2}(-2),$$

and the defining exact sequence is

$$0 \longrightarrow \mathcal{O}_{\mathbf{P}^2}(-2) \xrightarrow{\cdot(XZ-Y^2)} \mathcal{O}_{\mathbf{P}^2} \longrightarrow \mathcal{O}_C \longrightarrow 0.$$

At the same time, the projective morphism satisfies

$$\nu_2^* \mathcal{O}_{\mathbf{P}^2}(1) \cong \mathcal{O}_{\mathbf{P}^1}(2).$$

Thus this single example simultaneously illustrates:

- morphisms from projective space;
- homogeneous equations defining closed subschemes;
- coherent ideal sheaves;
- pullbacks of line bundles.

Scheme-theoretic inverse images and multiplicity

Example.

Let

$$g : \mathbf{A}_k^1 \longrightarrow \mathbf{A}_k^1, \quad u \longmapsto u^2.$$

Let

$$Z = V(t)$$

be the origin in the target affine line.

Set-theoretically,

$$g^{-1}(\{0\}) = \{0\}.$$

However, scheme-theoretically, the defining ideal pulls back by

$$(t) \longmapsto (u^2).$$

Therefore

$$g^{-1}(Z) = \operatorname{Spec} k[u]/(u^2).$$

Thus the scheme-theoretic inverse image is a double point.

Similarly, for the map

$$u \longmapsto u^3,$$

the inverse image of the origin is

$$\operatorname{Spec} k[u]/(u^3),$$

a triple point.

Hence:

scheme-theoretic inverse images remember multiplicity.

This is one of the most important reasons that schemes retain ideal-theoretic information rather than only underlying point sets.

9.7 Summary table of the examples

Example	Construction	Main conclusion
$\mathbf{A}_k^1$	$\operatorname{Spec} k[t]$	$\operatorname{Cl}(\mathbf{A}_k^1) = 0$
$\mathbf{A}_k^2$	$\operatorname{Spec} k[x, y]$	$\operatorname{Cl}(\mathbf{A}_k^2) = 0$
$\mathbf{P}_k^1$	Remove the point ∞	$\operatorname{Cl}(\mathbf{P}_k^1) \cong \mathbf{Z}$
$\mathbf{P}_k^1 \times \mathbf{P}_k^1$	Remove two ruling divisors	$\operatorname{Cl} \cong \mathbf{Z}^2$
$xy = z^2$	$\operatorname{div}(x) = 2D$	$\operatorname{Cl} \cong \mathbf{Z}/2\mathbf{Z}$
$xy = z^n$	$\operatorname{div}(x) = nD$	$\operatorname{Cl} \cong \mathbf{Z}/n\mathbf{Z}$
$xy = zw$	$\operatorname{div}(x) = D + E$	$\operatorname{Cl} \cong \mathbf{Z}$
Constant projective map	$[X : Y : Z] \mapsto [1 : 0 : 0 : 0]$	Image is a point
Linear embedding	$[X : Y : Z] \mapsto [X : Y : Z : 0]$	Image has dimension 2
Veronese conic	$[S : T] \mapsto [S^2 : ST : T^2]$	Image is $V(XZ - Y^2)$
Invalid projection	$[X : Y : Z] \dashrightarrow [X : Y]$	Not defined at $[0 : 0 : 1]$
Origin in $\mathbf{A}_k^2$	Ideal (x, y)	Reduced closed point
Double point	Ideal $(x^2) \subseteq k[x]$	Nonreduced thickening of one point
Fat point	Ideal $(x, y)^2 \subseteq k[x, y]$	First-order infinitesimal directions at the origin
Doubled line	Ideal $(y^2) \subseteq k[x, y]$	Nonreduced thickening of a line
Diagonal	Ideal $(x - y) \subseteq k[x, y]$	Closed copy of $\mathbf{A}_k^1$
Pullback under $u \mapsto u^2$	$(t) \mapsto (u^2)$	Reduced point pulls back to a double point
Noncoherent pushforward	$\mathbf{A}_k^1 \rightarrow \operatorname{Spec} k$	$p_*\mathcal{O}_{\mathbf{A}_k^1}$ is not coherent
Finite pushforward	$u \mapsto u^2$ on $\mathbf{A}_k^1$	$g_*\mathcal{O}_{\mathbf{A}_k^1}$ is coherent

9.8 Concluding observations

The divisor examples demonstrate that the divisor class group detects subtle failures of global principality.

For the surface cone

$$xy = z^2,$$

the relation

$$\operatorname{div}(x) = 2D$$

creates a nontrivial torsion class:

$$[D] \neq 0, \quad 2[D] = 0.$$

For the three-dimensional cone

$$xy = zw,$$

the relation

$$\operatorname{div}(x) = D + E$$

identifies two prime-divisor classes as opposites:

$$[E] = -[D],$$

leaving one free generator.

The projective examples show that a genuine morphism

$$\mathbf{P}_k^n \longrightarrow \mathbf{P}_k^m$$

must be represented by homogeneous polynomials with no common zero. The failed projection

$$[X : Y : Z] \dashrightarrow [X : Y]$$

is not a morphism precisely because of its base point. By contrast, the Veronese formula

$$[S : T] \longmapsto [S^2 : ST : T^2]$$

has no base point and gives a genuine closed embedding.

The sheaf examples show concretely that operations on quasi-coherent sheaves are module-theoretic on affine schemes. The exact sequence

$$0 \longrightarrow \mathcal{O}_{\mathbf{A}_k^1} \xrightarrow{t} \mathcal{O}_{\mathbf{A}_k^1} \longrightarrow \mathcal{O}_{\{0\}} \longrightarrow 0$$

is simply the sheaf version of

$$0 \longrightarrow k[t] \xrightarrow{t} k[t] \longrightarrow k[t]/(t) \longrightarrow 0.$$

Finally, the closed-subscheme examples explain why scheme theory is richer than ordinary point-set geometry. Although

$$V(x) = V(x^2)$$

as subsets of $\mathbf{A}_k^1$, the corresponding schemes

$$\operatorname{Spec} k[x]/(x) \quad \text{and} \quad \operatorname{Spec} k[x]/(x^2)$$

are not the same: the second contains a nilpotent infinitesimal direction.

Thus the examples support the central scheme-theoretic principle:

geometric spaces are understood through their local rings, modules, ideals, and rational functions.

10 Diagrams, Charts, and Summary Tables for Problems 4–7

This section collects visual summaries of the constructions appearing in the preceding problems. The central principle common to all of them is:

Global geometric structure is understood through algebra on affine open subsets.

The divisor-class computations reduce to factorial open subsets and units; morphisms from projective space reduce to homogeneous polynomials without common zeros; quasi-coherent sheaves reduce to modules; and closed subschemes reduce to ideals and quotient rings.

10.1 Global overview of the four themes

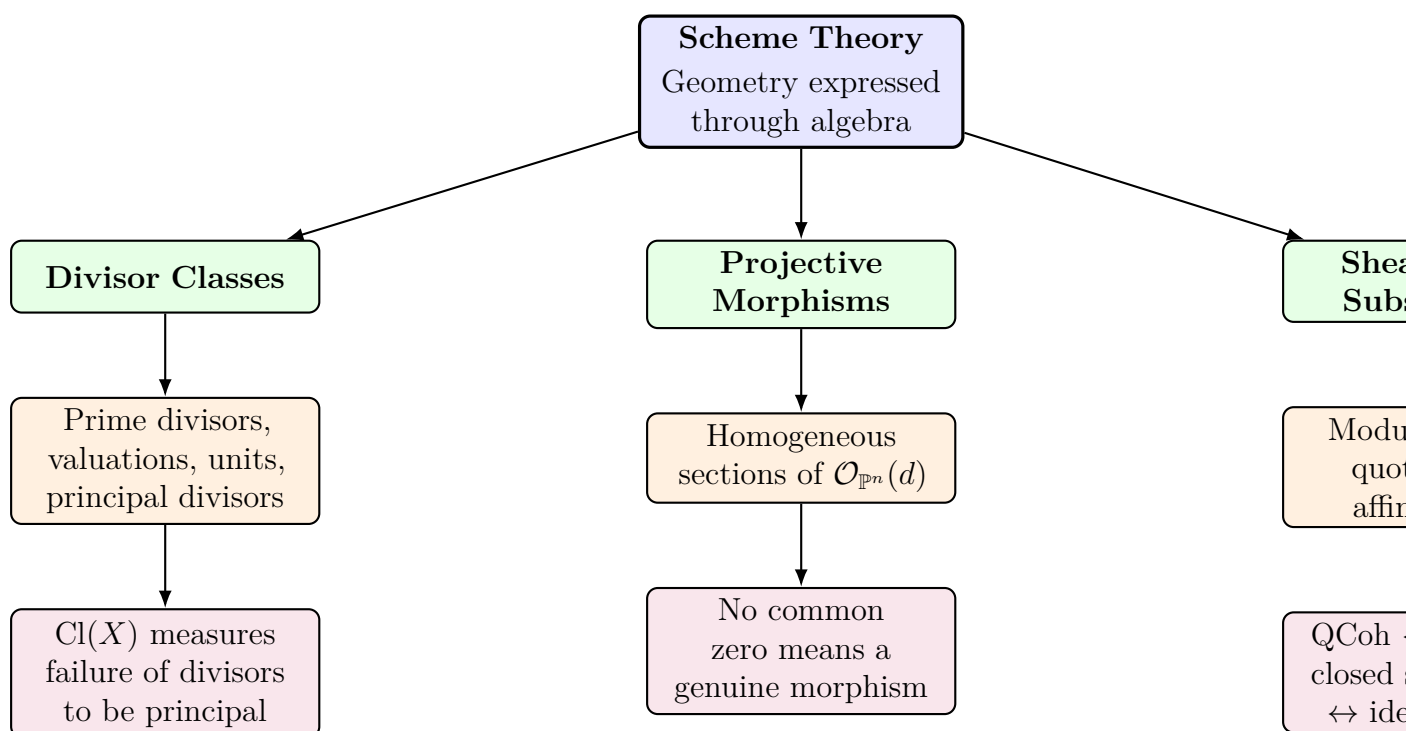


Figure 1: The four themes of Problems 4–7 and their local algebraic descriptions.

The diagram emphasizes the repeated pattern:

geometric object $\longrightarrow$ local algebraic model $\longrightarrow$ global geometric conclusion.

10.2 Divisor class groups: the general computation strategy

Flowchart for removing divisors

Let X be a normal integral Noetherian scheme. Suppose that

$$U = X \setminus (D_1 \cup \cdots \cup D_r),$$

where $D_1, \dots, D_r$ are the codimension-one irreducible components removed from X .

The divisor-class computation follows the scheme below.

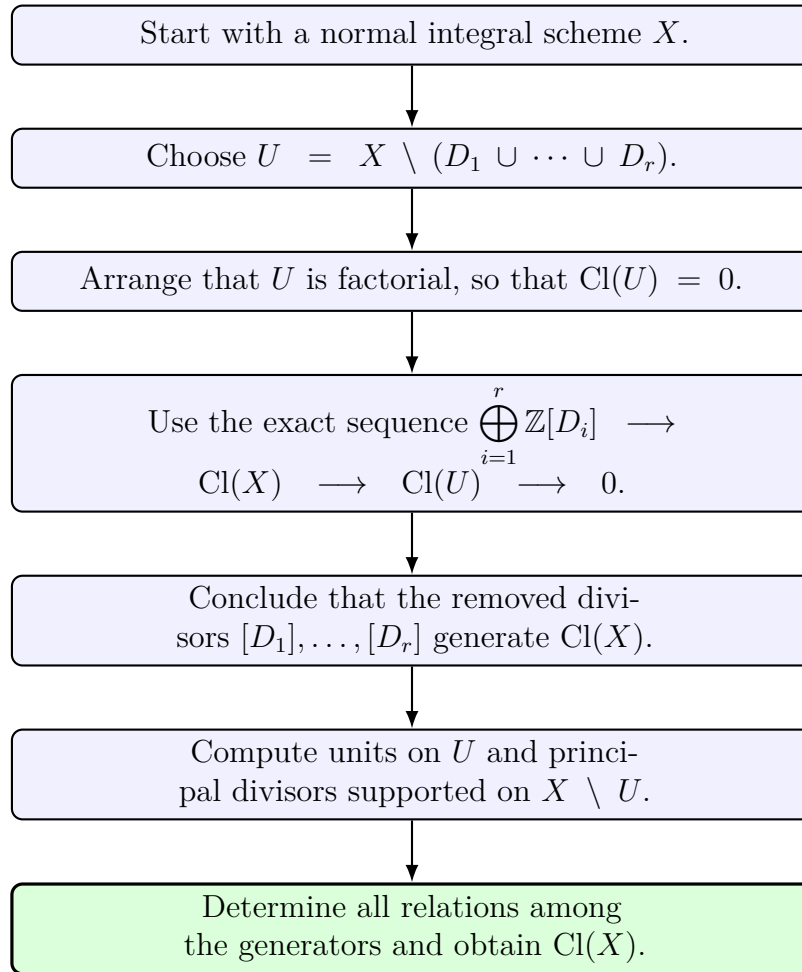


Figure 2: Standard strategy for computing divisor class groups by removing codimension-one subsets.

Units and possible relations

The possible relations among boundary divisors are controlled by units on the factorial open subset.

Open subset U	$\Gamma(U, \mathcal{O}_U)^\times$	Consequence for relations
$\text{Spec } k[s, t]$	$k^\times$	Only constant units; no nontrivial boundary relation
$\text{Spec } k[x, x^{-1}, z]$	$k^\times x^\mathbb{Z}$	Powers of x produce all relations
$\text{Spec } k[x, x^{-1}, z, w]$	$k^\times x^\mathbb{Z}$	Again, $\text{div}(x)$ controls all relations

10.3 Comparison chart for the three class-group computations

Scheme X	Factorial open subset U	Removed prime divisors	Principal relation	Result
$\mathbb{P}_k^1 \times \mathbb{P}_k^1$	$\mathbb{A}_k^2 = \text{Spec } k[s, t]$	$D_1 = \{\infty\} \times \mathbb{P}^1, \quad D_2 = \mathbb{P}^1 \times \{\infty\}$	No nontrivial relation	$\text{Cl}(X) \cong \mathbb{Z}^2$
$\text{Spec } k[x, y, z]/(xy - z^2)$	$\text{Spec } k[x, x^{-1}, z]$	$D = V(x, z)$	$\text{div}(x) = 2D$	$\text{Cl}(X) \cong \mathbb{Z}/2\mathbb{Z}$
$\text{Spec } k[x, y, z, w]/(xy - zw)$	$\text{Spec } k[x, x^{-1}, z, w]$	$D = V(x, z), \quad E = V(x, w)$	$\text{div}(x) = D + E$	$\text{Cl}(X) \cong \mathbb{Z}$

Table 1: The three divisor-class computations from Problem 4.

10.4 The rulings of $\mathbb{P}_k^1 \times \mathbb{P}_k^1$

Geometric ruling diagram

The product surface

$$\mathbb{P}_k^1 \times \mathbb{P}_k^1$$

contains two independent families of curves:

$$\{p\} \times \mathbb{P}_k^1 \quad \text{and} \quad \mathbb{P}_k^1 \times \{q\}.$$

Removing the boundary divisors

The lattice of divisor classes

Every divisor class on $\mathbb{P}_k^1 \times \mathbb{P}_k^1$ can be written as

$$(a, b) = a[D_1] + b[D_2].$$

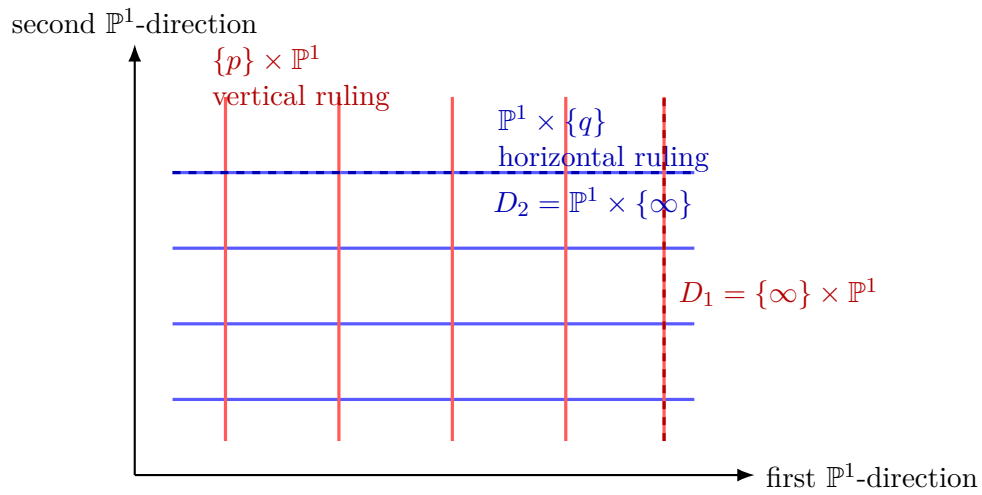
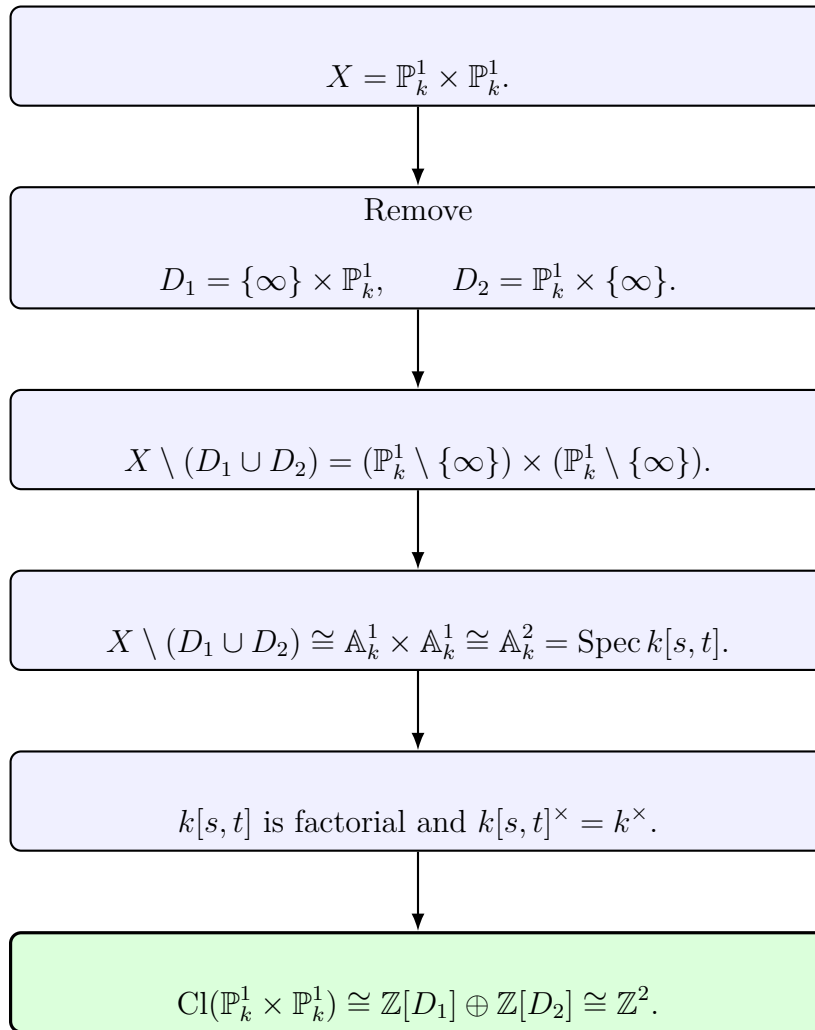
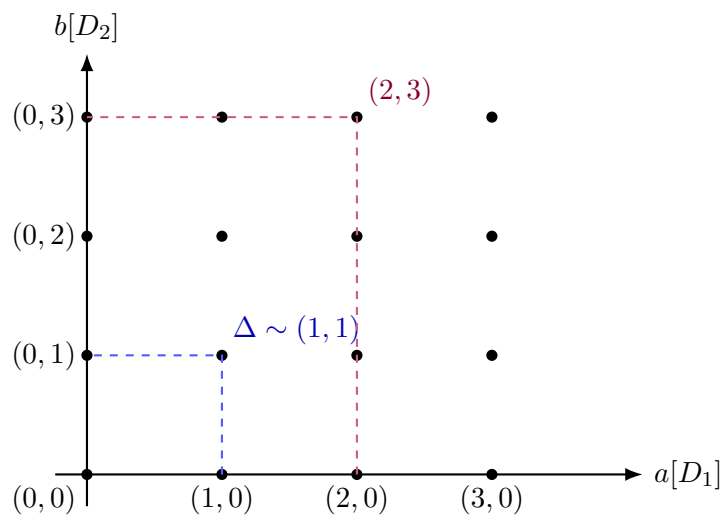
Figure 3: The two independent ruling families on $\mathbb{P}_k^1 \times \mathbb{P}_k^1$.

Figure 4: Removal of two boundary rulings reduces the class-group computation to the factorial affine plane.


 Figure 5: The divisor-class lattice $\text{Cl}(\mathbb{P}_k^1 \times \mathbb{P}_k^1) \cong \mathbb{Z}^2$.

Divisor on $\mathbb{P}_k^1 \times \mathbb{P}_k^1$	Class
$\{p\} \times \mathbb{P}_k^1$	$(1, 0)$
$\mathbb{P}_k^1 \times \{q\}$	$(0, 1)$
Diagonal $\Delta = \{([S_0 : S_1], [T_0 : T_1]) : S_0 T_1 - S_1 T_0 = 0\}$	$(1, 1)$
A divisor given by a bihomogeneous equation of degree (a, b)	(a, b)

 Table 2: Selected divisor classes on $\mathbb{P}_k^1 \times \mathbb{P}_k^1$.

10.5 The quadric cone $xy = z^2$

The class-group mechanism

Let

$$X = \operatorname{Spec} k[x, y, z]/(xy - z^2).$$

The relevant prime divisor is

$$D = V(x, z).$$

Multiplicity interpretation

Although the reduced support of $V(x)$ is the line

$$D = V(x, z),$$

the equation $xy = z^2$ implies that x vanishes to order 2 along this line.

$$V(x) \subseteq X$$

$$\cup$$

scheme-theoretically defined by $A/(x) \cong k[y, z]/(z^2)$

$$\Downarrow$$

$$V(x)_{\text{red}} = V(x, z) = D \quad \text{with multiplicity 2}$$

$$\Downarrow$$

$$\operatorname{div}(x) = 2D$$

$$\Downarrow$$

$$[D] \text{ has order 2 in } \operatorname{Cl}(X).$$

The family $xy = z^n$

The same argument works for the entire family

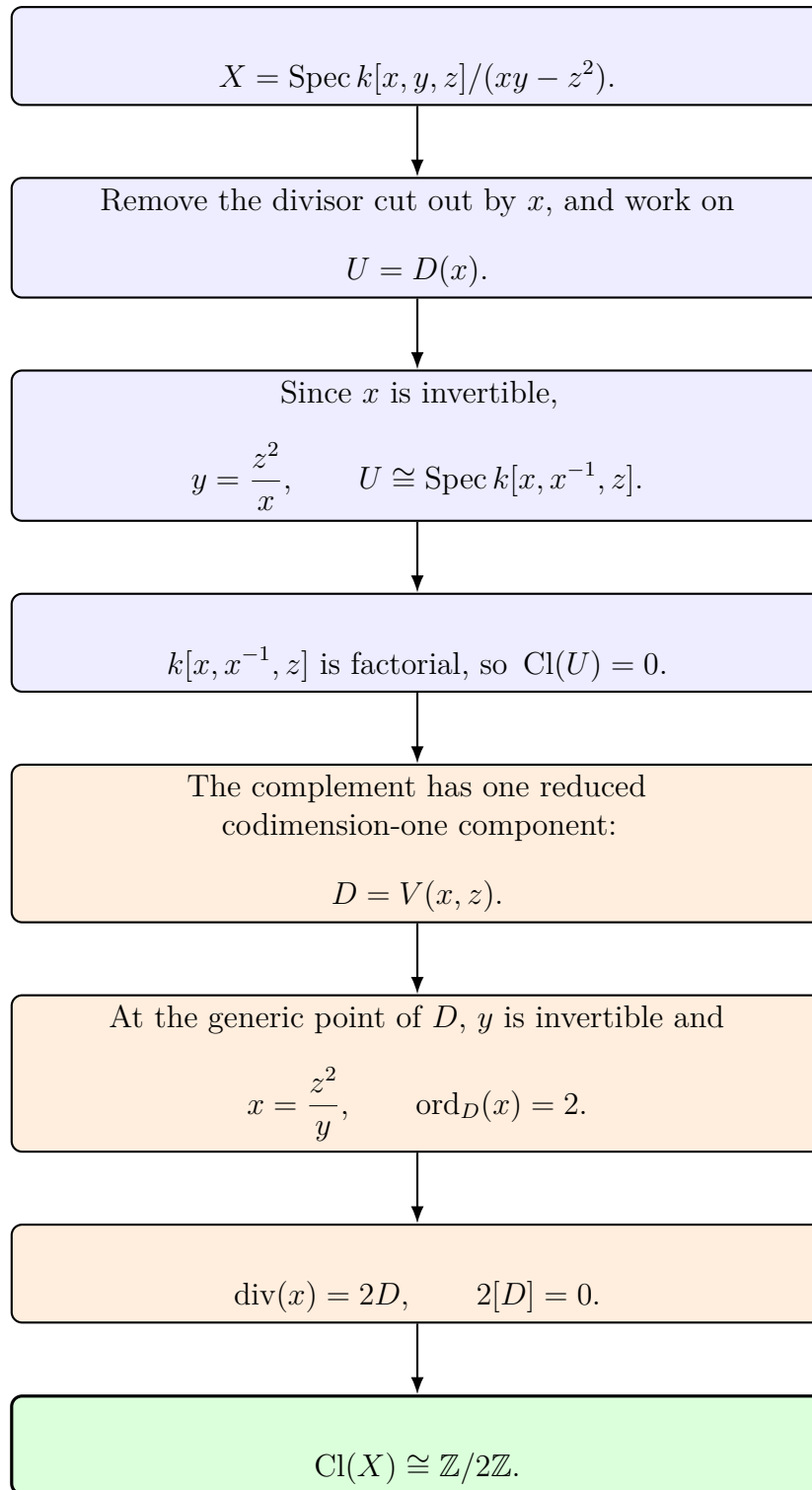
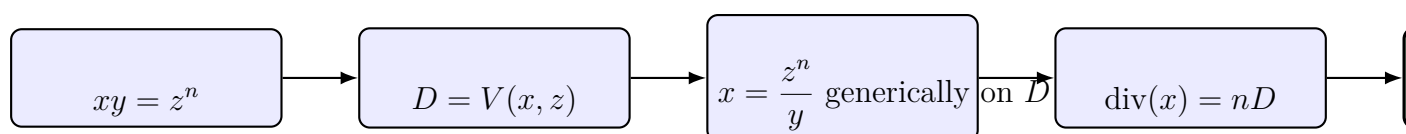
$$X_n = \operatorname{Spec} k[x, y, z]/(xy - z^n).$$

10.6 The three-dimensional quadric cone $xy = zw$

Boundary decomposition

Let

$$X = \operatorname{Spec} k[x, y, z, w]/(xy - zw).$$

Figure 6: Why the divisor class group of the quadric surface cone is $\mathbb{Z}/2\mathbb{Z}$.Figure 7: The cyclic class groups of the family $xy = z^n$.

Equation	Principal relation	Class group
$xy = z^2$	$\operatorname{div}(x) = 2D$	$\mathbb{Z}/2\mathbb{Z}$
$xy = z^3$	$\operatorname{div}(x) = 3D$	$\mathbb{Z}/3\mathbb{Z}$
$xy = z^4$	$\operatorname{div}(x) = 4D$	$\mathbb{Z}/4\mathbb{Z}$
$xy = z^n$	$\operatorname{div}(x) = nD$	$\mathbb{Z}/n\mathbb{Z}$

Table 3: Torsion class groups arising from multiplicity n along one boundary divisor.

Removing $V(x)$ gives a factorial open subset. Unlike the cone $xy = z^2$, the boundary now splits into two distinct prime divisors.

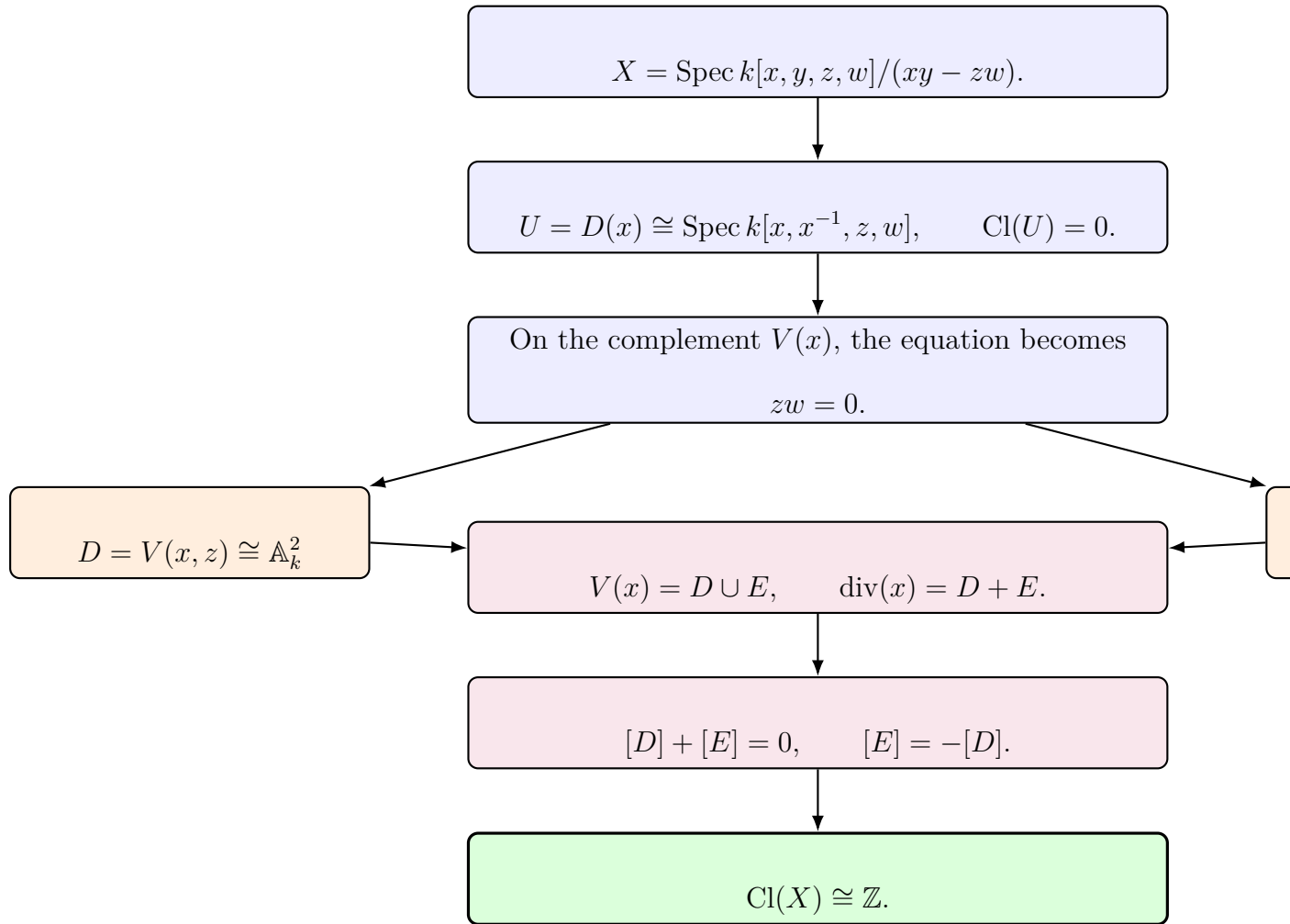


Figure 8: Boundary decomposition and divisor-class relation for the cone $xy = zw$.

Feature	$xy = z^2$	$xy = zw$
Dimension of the affine cone	2	3
Boundary after removing $D(x)$	One prime divisor $D = V(x, z)$	Two prime divisors $D = V(x, z), \quad E = V(x, w)$
Structure of $V(x)$	One component with multiplicity 2	Two distinct components, each with multiplicity 1
Principal divisor of x	$\operatorname{div}(x) = 2D$	$\operatorname{div}(x) = D + E$
Relation in the class group	$2[D] = 0$	$[D] + [E] = 0$
Remaining class behavior	Nonzero torsion class	One free generator
Divisor class group	$\operatorname{Cl}(X) \cong \mathbb{Z}/2\mathbb{Z}$	$\operatorname{Cl}(X) \cong \mathbb{Z}$

Table 4: Multiplicity versus splitting in the two quadric cone examples.

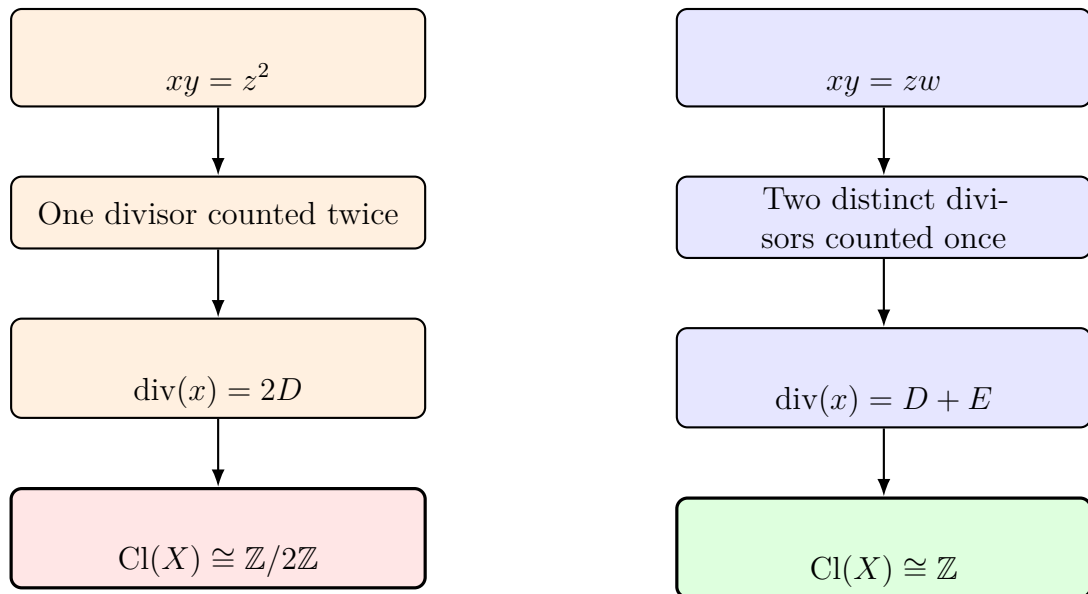


Figure 9: One divisor with multiplicity versus two distinct divisor components.

Comparison of the two cones

10.7 Morphisms from projective space

Morphisms as homogeneous coordinate systems

A morphism

$$\varphi : \mathbb{P}_k^n \longrightarrow \mathbb{P}_k^m$$

is represented by homogeneous forms of the same degree:

$$\varphi([X_0 : \cdots : X_n]) = [F_0(X) : \cdots : F_m(X)].$$

The forms must have no common projective zero.

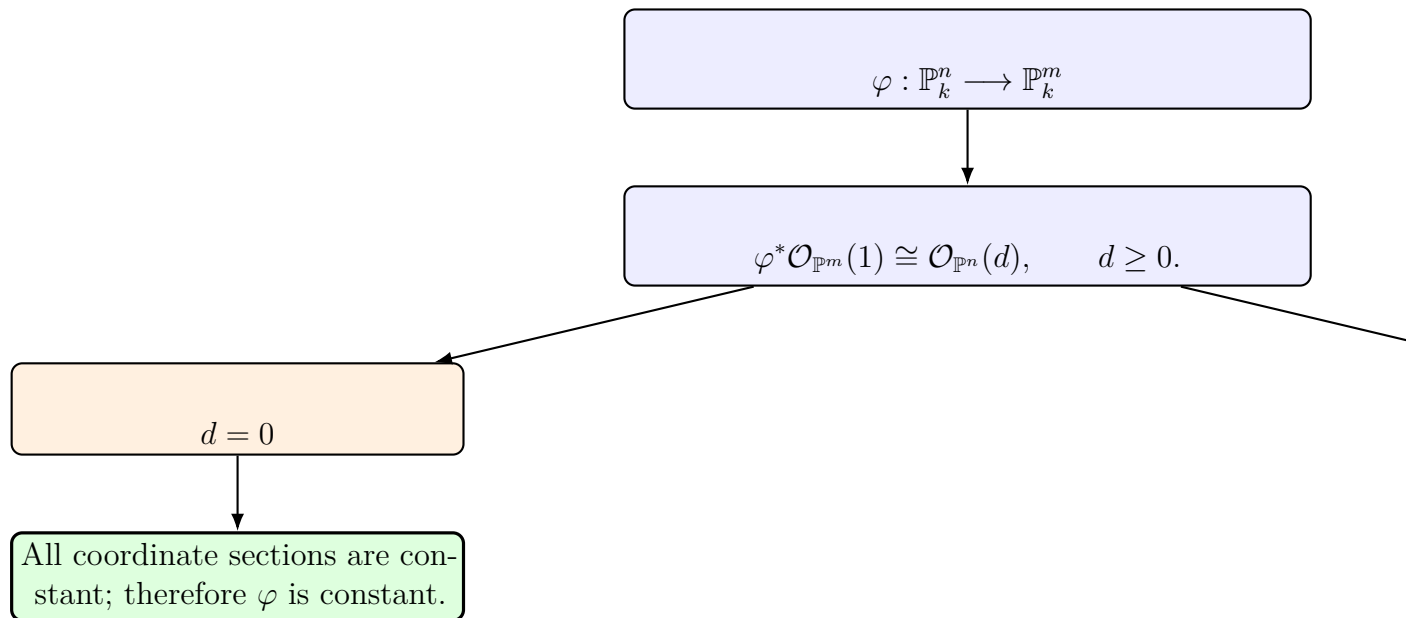


Figure 10: The line-bundle description of morphisms from projective space.

Valid and invalid projective coordinate formulas

The failed projection from $\mathbb{P}^2$ to $\mathbb{P}^1$

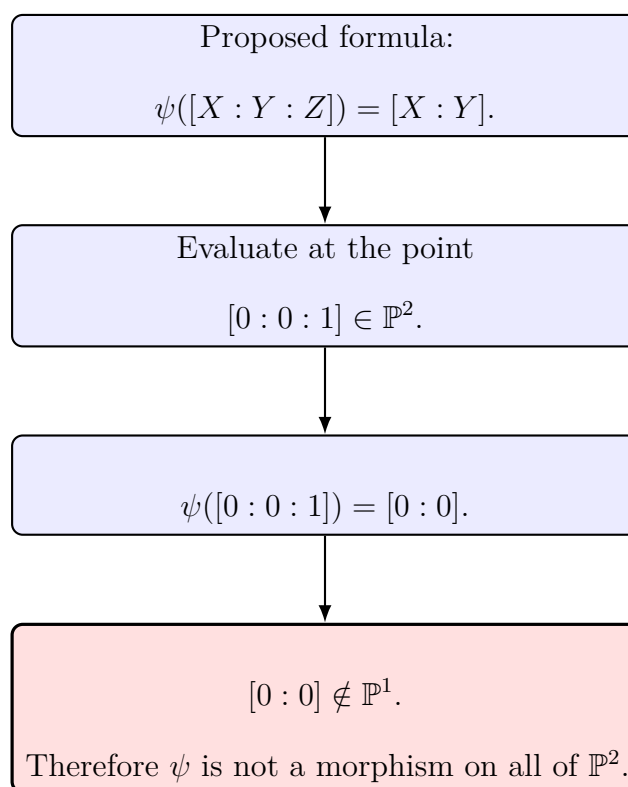
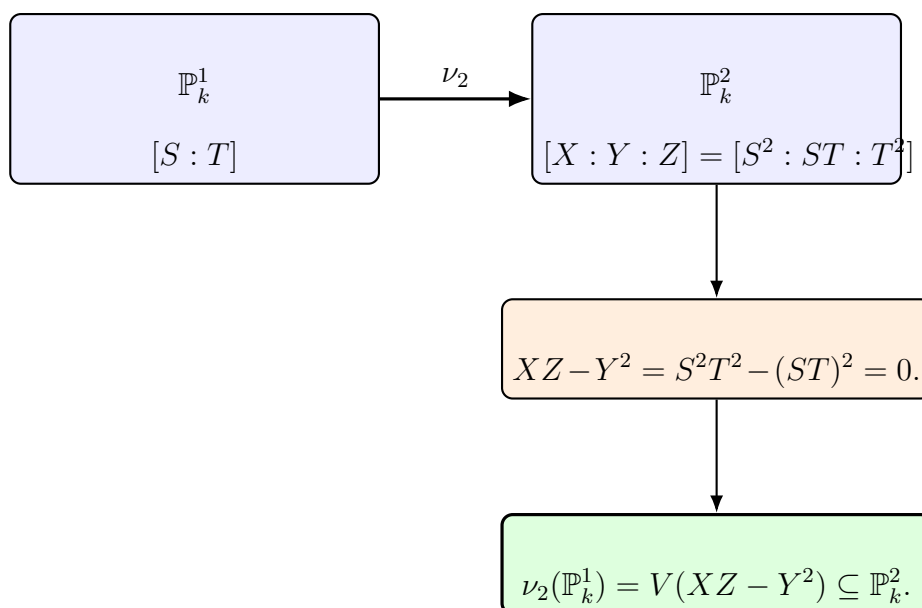
The Veronese conic

The quadratic Veronese morphism is

$$\nu_2 : \mathbb{P}_k^1 \longrightarrow \mathbb{P}_k^2, \quad [S : T] \longmapsto [S^2 : ST : T^2].$$

Formula	Source and target	Morphism?	Reason
$[X : Y : Z] \mapsto [1 : 0 : 0 : 0]$	$\mathbb{P}^2 \rightarrow \mathbb{P}^3$	Yes	Constant morphism; the target tuple is always defined.
$[X : Y : Z] \mapsto [X : Y : Z : 0]$	$\mathbb{P}^2 \rightarrow \mathbb{P}^3$	Yes	Linear embedding; the first three coordinates do not vanish simultaneously.
$[S : T] \mapsto [S^2 : ST : T^2]$	$\mathbb{P}^1 \rightarrow \mathbb{P}^2$	Yes	The three quadratic forms have no common projective zero.
$[X : Y : Z] \mapsto [X^2 : Y^2 : Z^2]$	$\mathbb{P}^2 \rightarrow \mathbb{P}^2$	Yes	The three forms vanish simultaneously only at the forbidden affine origin.
$[X : Y : Z] \mapsto [X : Y]$	$\mathbb{P}^2 \dashrightarrow \mathbb{P}^1$	No	Undefined at $[0 : 0 : 1]$.
$[X : Y : Z] \mapsto [X^2 : XY : Y^2]$	$\mathbb{P}^2 \dashrightarrow \mathbb{P}^2$	No	All three coordinates vanish at $[0 : 0 : 1]$.

Table 5: Examples of genuine morphisms and rational maps with base points.

Figure 11: Why ordinary projection gives only a rational map $\mathbb{P}^2 \dashrightarrow \mathbb{P}^1$.Figure 12: The Veronese embedding realizes $\mathbb{P}_k^1$ as a smooth plane conic.

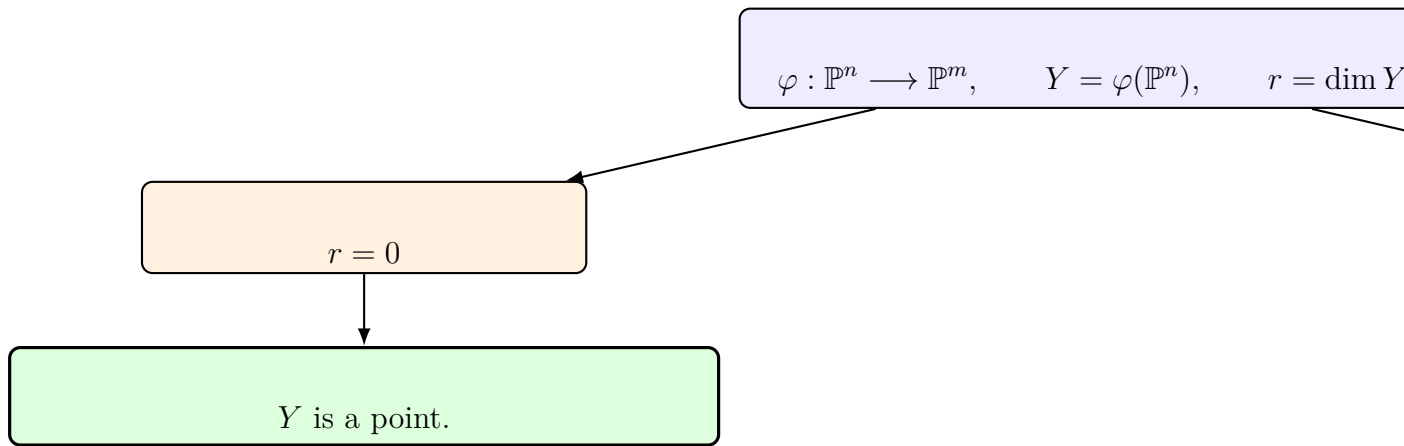


Figure 13: The dimension argument for morphisms from projective space.

Feature of the Veronese morphism	Value
Source	$\mathbb{P}_k^1$
Target	$\mathbb{P}_k^2$
Formula	$[S : T] \mapsto [S^2 : ST : T^2]$
Degree of coordinate forms	2
Pullback of the hyperplane bundle	$\nu_2^* \mathcal{O}_{\mathbb{P}^2}(1) \cong \mathcal{O}_{\mathbb{P}^1}(2)$
Equation of the image	$XZ - Y^2 = 0$
Dimension of the image	$1 = \dim \mathbb{P}_k^1$

Table 6: Basic data for the quadratic Veronese embedding.

Dimension-rigidity flowchart

10.8 Quasi-coherent and coherent sheaves

The affine dictionary

Let

$$X = \operatorname{Spec} A.$$

The correspondence between modules and quasi-coherent sheaves can be summarized as follows.

Exactness under sheafification

Multiplication by t on the affine line

Let

$$X = \mathbb{A}_k^1 = \operatorname{Spec} k[t].$$

The module exact sequence

$$0 \longrightarrow k[t] \xrightarrow{\cdot t} k[t] \longrightarrow k[t]/(t) \longrightarrow 0$$

gives the sheaf exact sequence

$$0 \longrightarrow \mathcal{O}_{\mathbb{A}_k^1} \xrightarrow{\cdot t} \mathcal{O}_{\mathbb{A}_k^1} \longrightarrow \mathcal{O}_{\{0\}} \longrightarrow 0.$$

Algebra over A	Geometry on $X = \operatorname{Spec} A$
A -module M	Quasi-coherent sheaf $\widetilde{M}$
The ring A	Structure sheaf $\widetilde{A} = \mathcal{O}_X$
Ideal $I \subseteq A$	Ideal sheaf $\widetilde{I} \subseteq \mathcal{O}_X$
Quotient A/I	Structure sheaf of the closed subscheme $\operatorname{Spec}(A/I) \hookrightarrow \operatorname{Spec} A$
Homomorphism $\phi : M \rightarrow N$	Morphism $\widetilde{\phi} : \widetilde{M} \rightarrow \widetilde{N}$
$\ker \phi$	$\ker(\widetilde{\phi}) \cong \widetilde{\ker \phi}$
$\operatorname{im} \phi$	$\operatorname{im}(\widetilde{\phi}) \cong \widetilde{\operatorname{im} \phi}$
$\operatorname{coker} \phi$	$\operatorname{coker}(\widetilde{\phi}) \cong \widetilde{\operatorname{coker} \phi}$
Tensor product $M \otimes_A B$	Pullback under $\operatorname{Spec} B \longrightarrow \operatorname{Spec} A$

Table 7: The affine correspondence between algebraic operations and sheaf-theoretic constructions.

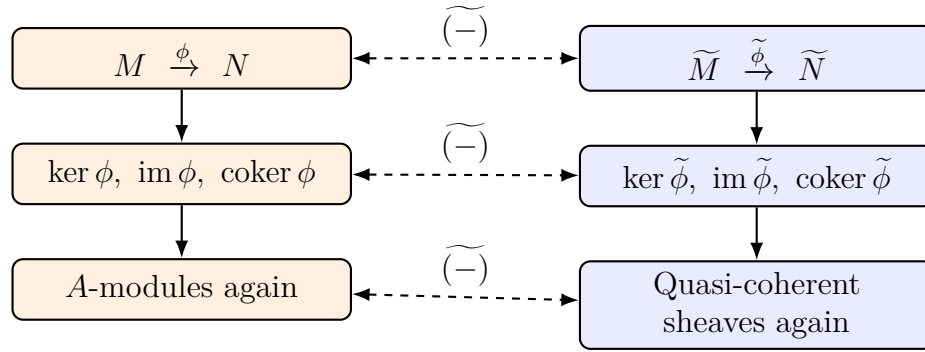


Figure 14: Kernels, images, and cokernels of module maps correspond to those of quasi-coherent sheaf morphisms on affine schemes.

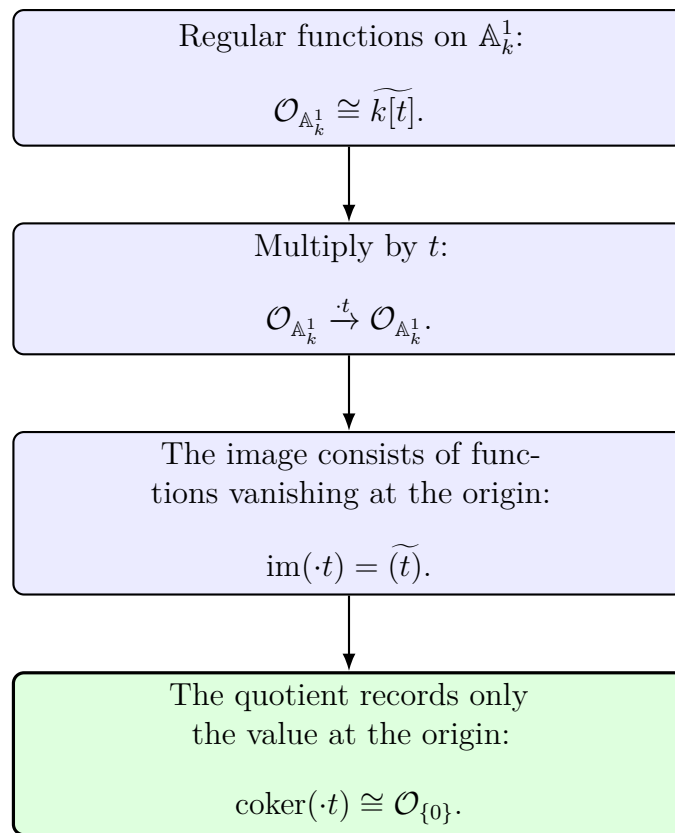


Figure 15: The sheaf-theoretic meaning of multiplication by t on $\mathbb{A}_k^1$.

Part of the map	Module form	Sheaf form
Source	$k[t]$	$\mathcal{O}_{\mathbb{A}_k^1}$
Map	Multiplication by t	Multiplication by t
Kernel	0	0
Image	(t)	$(\widetilde{t})$
Cokernel	$k[t]/(t) \cong k$	$\mathcal{O}_{\{0\}}$

Table 8: The exact sequence associated to the origin of the affine line.

A coherent resolution of the origin in $\mathbb{A}_k^2$

The origin of

$$\mathbb{A}_k^2 = \operatorname{Spec} k[x, y]$$

is defined by the ideal (x, y) . The structure sheaf of the origin admits the finite free resolution

$$0 \longrightarrow \mathcal{O}_{\mathbb{A}_k^2} \xrightarrow{\begin{pmatrix} -y \\ x \end{pmatrix}} \mathcal{O}_{\mathbb{A}_k^2}^{\oplus 2} \xrightarrow{\begin{pmatrix} x & y \end{pmatrix}} \mathcal{O}_{\mathbb{A}_k^2} \longrightarrow \mathcal{O}_{\{(0,0)\}} \longrightarrow 0.$$

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10.9 Pullbacks and pushforwards

Pullback as extension of scalars

Suppose

$$g : \operatorname{Spec} B \longrightarrow \operatorname{Spec} A$$

corresponds to a ring map

$$A \longrightarrow B.$$

For an A -module M ,

$$g^*(\widetilde{M}) \cong \widetilde{M \otimes_A B}.$$

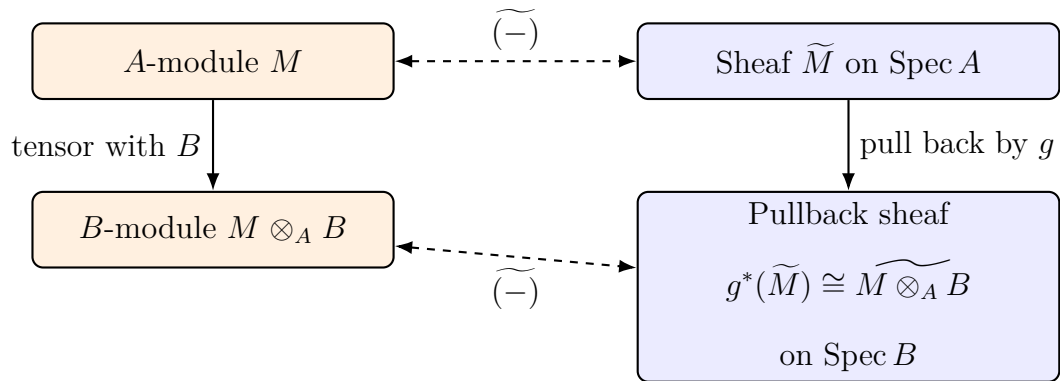


Figure 16: Pullback of quasi-coherent sheaves is extension of scalars on affine schemes.

Pulling back a point under the squaring map

Let

$$g : \mathbb{A}_u^1 \longrightarrow \mathbb{A}_t^1, \quad t \longmapsto u^2.$$

The reduced origin in the target is defined by the ideal

$$(t).$$

Its pullback is defined by

$$(t) \longmapsto (u^2).$$

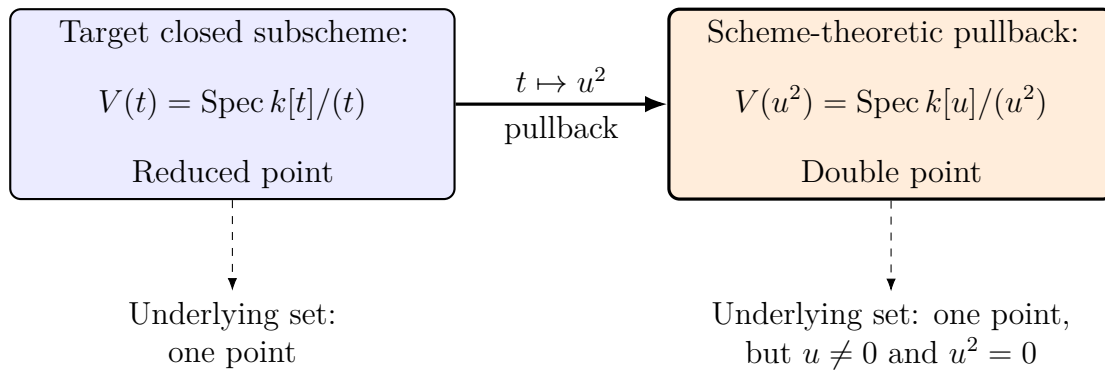


Figure 17: A reduced point pulls back to a double point under $u \mapsto u^2$.

Multiplicity chart for fibers over the origin

Pushforward: failure and preservation of coherence

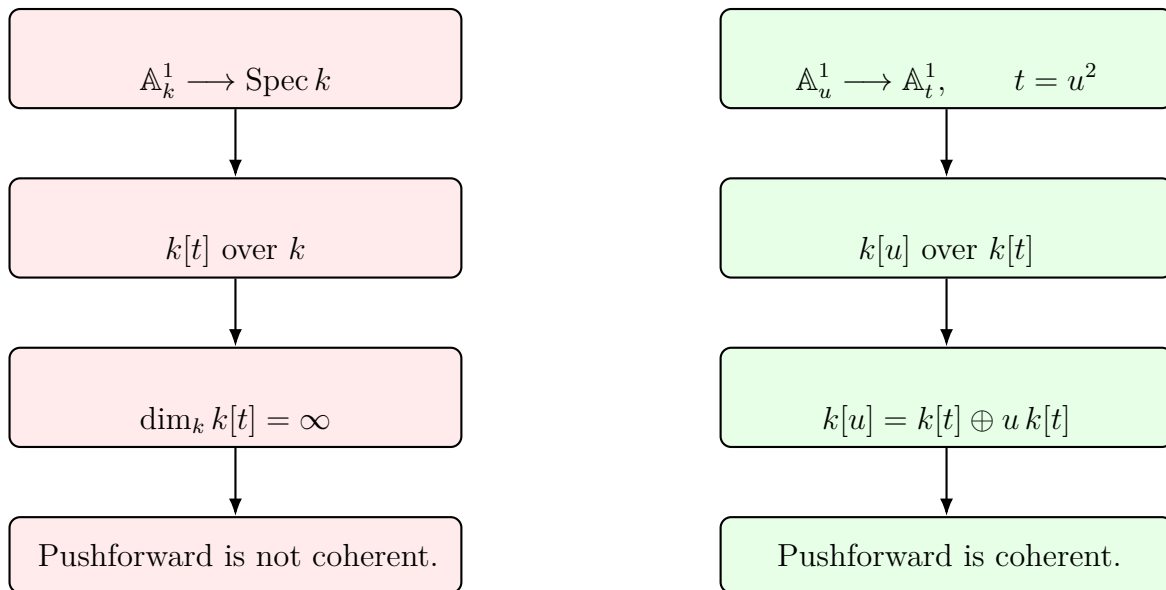


Figure 18: Negative and positive pushforward examples for coherence.

10.10 Closed subschemes and ideal sheaves

Closed sets versus closed subschemes

Let

$$X = \operatorname{Spec} A.$$

Morphism $g : \mathbb{A}_u^1 \rightarrow \mathbb{A}_t^1$	Pulled-back ideal	Scheme-theoretic fiber over 0
$t \mapsto u$	(u)	Reduced point $\text{Spec } k[u]/(u)$
$t \mapsto u^2$	(u^2)	Double point $\text{Spec } k[u]/(u^2)$
$t \mapsto u^3$	(u^3)	Triple point $\text{Spec } k[u]/(u^3)$
$t \mapsto u^n$	(u^n)	Point of multiplicity n $\text{Spec } k[u]/(u^n)$

Table 9: Scheme-theoretic fibers preserve the multiplicity of vanishing.

Morphism	Pushforward of the structure sheaf	Finite on the target?	Coherent?
$p : \mathbb{A}_k^1 \rightarrow \text{Spec } k$	$p_* \mathcal{O}_{\mathbb{A}_k^1} \hookrightarrow k[t]$ as a k -vector space	No; $k[t]$ is infinite-dimensional over k	No
$g : \mathbb{A}_u^1 \rightarrow \mathbb{A}_t^1, \quad t = u^2$	$g_* \mathcal{O}_{\mathbb{A}_u^1} \hookrightarrow k[u]$ as a $k[t]$ -module	Yes; $k[u] = k[t] \oplus u k[t]$	Yes

Table 10: Pushforward does not preserve coherence in general, but finite morphisms behave well.

The ideal $I \subseteq A$ determines the closed subscheme

$$\mathrm{Spec}(A/I) \hookrightarrow \mathrm{Spec} A,$$

whereas its radical $\sqrt{I}$ determines only the underlying closed set.

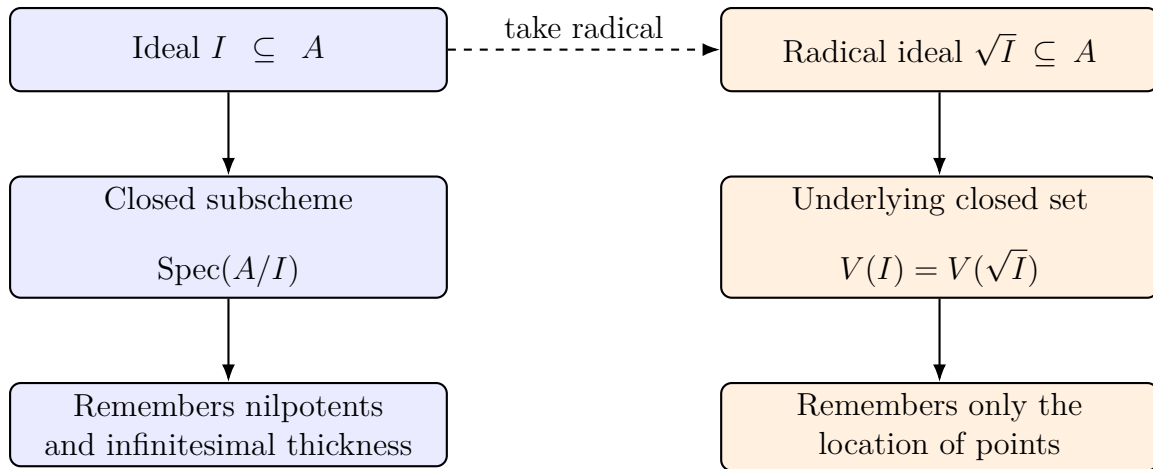


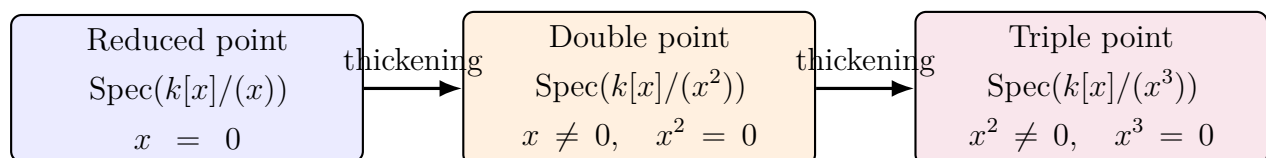
Figure 19: A closed subscheme contains more information than its underlying closed subset.

Infinitesimal thickenings of a point

The ideals

$$(x), \quad (x^2), \quad (x^3)$$

all define the same underlying topological point in $\mathbb{A}_k^1$, but they define different closed subschemes.



$$V(x) = V(x^2) = V(x^3) = \{0\}$$

as underlying topological spaces.

Figure 20: Different scheme structures supported on the same point of $\mathbb{A}_k^1$.

Closed subschemes of the affine plane

Schematic drawings in $\mathbb{A}_k^2$

Ideal sheaves and closed subschemes

A closed immersion

$$i : Z \hookrightarrow X$$

Ideal	Coordinate ring	k-vector-space length	Interpretation
(x)	$k[x]/(x) \cong k$	1	Reduced point
(x^2)	$k[x]/(x^2)$	2	Double point
(x^3)	$k[x]/(x^3)$	3	Triple point
(x^n)	$k[x]/(x^n)$	n	Point thickened to order n

Table 11: Infinitesimal thickenings supported at the origin of $\mathbb{A}_k^1$.

Ideal in $k[x, y]$	Underlying closed set	Scheme-theoretic meaning
(x, y)	The origin	Reduced closed point
$(x, y)^2 = (x^2, xy, y^2)$	The origin	Fat point retaining first-order x - and y -directions
(y)	The x -axis	Reduced affine line
(y^2)	The x -axis	Doubled line with nilpotent normal direction
(xy)	Union of the coordinate axes	Reduced reducible crossing, singu- lar at the origin
$(x - y)$	The diagonal line	Closed copy of $\mathbb{A}_k^1$
$(y - x^2)$	The parabola $y = x^2$	Graph of the morphism $x \mapsto x^2$

Table 12: Several closed subschemes of $\mathbb{A}_k^2$.

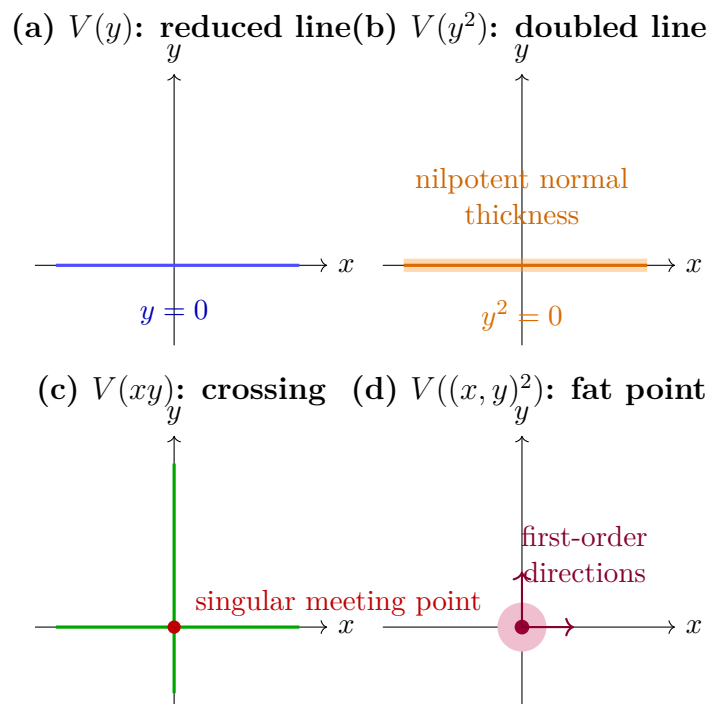


Figure 21: Visual representations of selected closed subschemes of $\mathbb{A}_k^2$. The thickness in panels (b) and (d) represents nilpotent scheme structure rather than additional topological points.

is represented by its quasi-coherent ideal sheaf

$$\mathcal{I}_{Z/X} = \ker(\mathcal{O}_X \rightarrow i_*\mathcal{O}_Z).$$

Ideal sheaves and closed subschemes

A closed immersion

$$i : Z \hookrightarrow X$$

is encoded by a quasi-coherent ideal sheaf. Equivalently, one has the fundamental exact sequence

$$0 \longrightarrow \mathcal{I}_{Z/X} \longrightarrow \mathcal{O}_X \longrightarrow i_*\mathcal{O}_Z \longrightarrow 0.$$

Conversely, a quasi-coherent ideal sheaf reconstructs a closed subscheme locally and then globally.

10.11 Projective hypersurfaces as closed subschemes

A hyperplane in projective space

Let

$$H = V(X_0) \subseteq \mathbb{P}_k^n.$$

Then

$$H \cong \mathbb{P}_k^{n-1},$$

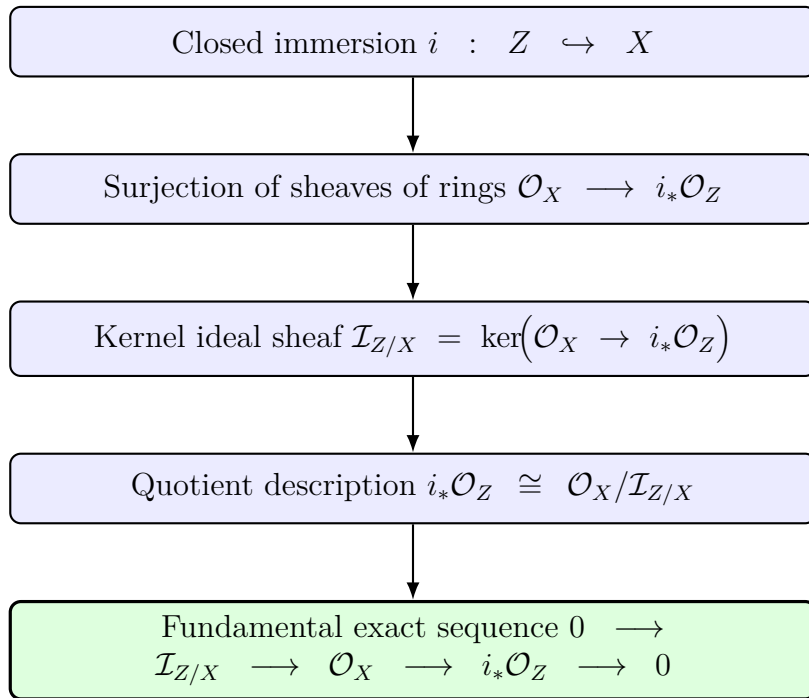


Figure 22: A closed subscheme is encoded by its quasi-coherent ideal sheaf.

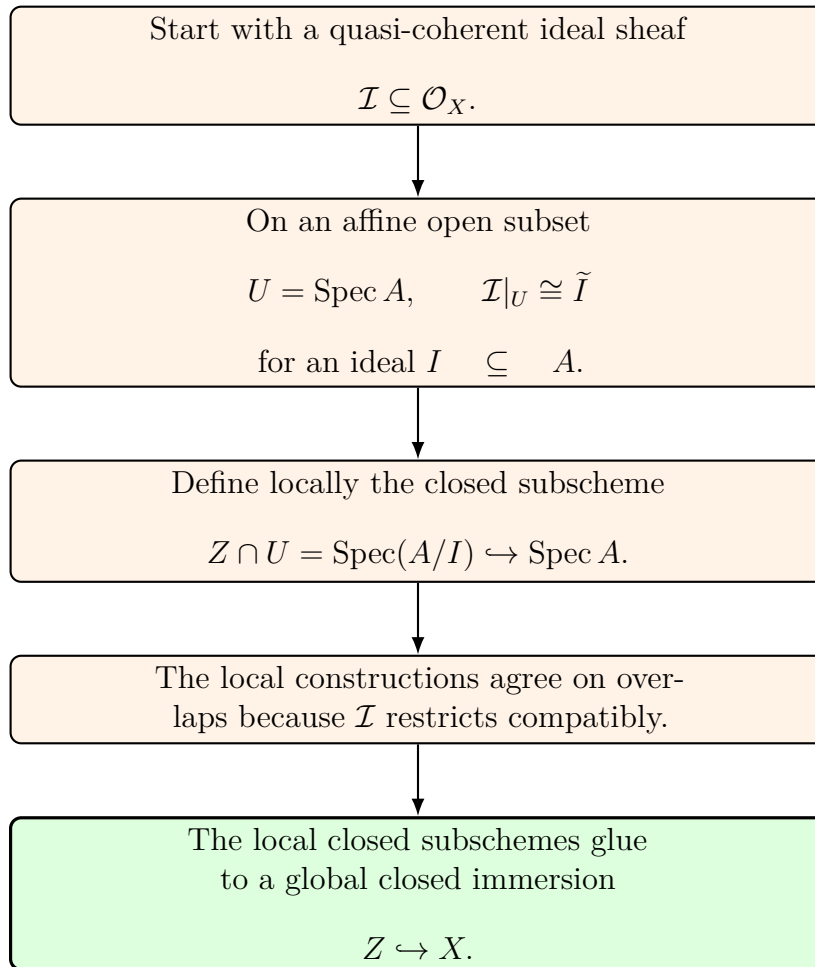


Figure 23: Construction of a closed subscheme from a quasi-coherent ideal sheaf.

and its ideal sheaf is

$$\mathcal{I}_H \cong \mathcal{O}_{\mathbb{P}^n}(-1).$$

$$0 \longrightarrow \boxed{\mathcal{O}_{\mathbb{P}^n}(-1)} \xrightarrow{\cdot X_0} \boxed{\mathcal{O}_{\mathbb{P}^n}} \longrightarrow \boxed{\mathcal{O}_H} \longrightarrow 0$$

Figure 24: The defining exact sequence of a hyperplane $H \subseteq \mathbb{P}_k^n$.

Thus:

$$\boxed{0 \longrightarrow \mathcal{O}_{\mathbb{P}^n}(-1) \xrightarrow{\cdot X_0} \mathcal{O}_{\mathbb{P}^n} \longrightarrow \mathcal{O}_H \longrightarrow 0.}$$

The Veronese conic as a closed subscheme

The conic

$$C = V(XZ - Y^2) \subseteq \mathbb{P}_k^2$$

is defined by a quadratic equation. Hence

$$\mathcal{I}_C \cong \mathcal{O}_{\mathbb{P}^2}(-2).$$

$$\boxed{\mathbb{P}_k^1} \xrightarrow{\nu_2} \boxed{\mathbb{P}_k^2} \quad [S : T] \mapsto [S^2 : ST : T^2]$$

$$\boxed{\mathcal{O}_{\mathbb{P}^2}(-2)} \xrightarrow{\cdot (XZ - Y^2)} \boxed{\mathcal{O}_{\mathbb{P}^2}} \longrightarrow \boxed{\mathcal{O}_C} \longrightarrow 0$$

Figure 25: The Veronese conic as both the image of a morphism and a closed subscheme defined by an ideal sheaf.

Thus:

$$\boxed{0 \longrightarrow \mathcal{O}_{\mathbb{P}^2}(-2) \xrightarrow{\cdot (XZ - Y^2)} \mathcal{O}_{\mathbb{P}^2} \longrightarrow \mathcal{O}_C \longrightarrow 0.}$$

10.12 Examples linking the four topics

The divisor $D = V(x, z)$ on $xy = z^2$

Let

$$X = \operatorname{Spec} k[x, y, z]/(xy - z^2).$$

The closed subscheme

$$D = V(x, z)$$

connects the theory of ideal sheaves with the theory of divisor classes.

Viewpoint	Description of $D = V(x, z)$
Closed subscheme	Defined by the ideal $(x, z) \subseteq k[x, y, z]/(xy - z^2).$
Coordinate ring	$\frac{k[x, y, z]/(xy - z^2)}{(x, z)} \cong k[y].$ Therefore $D \cong \mathbb{A}_k^1.$
Prime divisor	D is irreducible and codimension one in the surface X .
Divisor-class relation	$\operatorname{div}(x) = 2D.$
Class-group meaning	$[D] \neq 0, \quad 2[D] = 0, \quad \operatorname{Cl}(X) \cong \mathbb{Z}/2\mathbb{Z}.$

Table 13: The ruling line on the surface cone viewed simultaneously as a closed subscheme and a nontrivial divisor class.

The divisors D and E on $xy = zw$

Let

$$X = \operatorname{Spec} k[x, y, z, w]/(xy - zw).$$

“ $\text{\LaTeX}$

Feature	$D = V(x, z)$	$E = V(x, w)$
Defining ideal	(x, z)	(x, w)
Coordinate ring	$A/(x, z) \cong k[y, w]$	$A/(x, w) \cong k[y, z]$
Isomorphism type	$D \cong \mathbb{A}_k^2$	$E \cong \mathbb{A}_k^2$
Divisor relation	$\operatorname{div}(x) = D + E$	
Class-group relation	$[D] + [E] = 0, \quad [E] = -[D]$	

Table 14: The two prime boundary divisors on the three-dimensional quadric cone.

10.13 Complete synthesis table

““

Theme	Local algebra	Central relation or construction	Geometric conclusion
Divisor class groups	Height-one primes, valuations, and units	$\mathrm{Cl}(X) = \mathrm{Div}(X)/\mathrm{Prin}(X)$	Measures the failure of codimension-one subvarieties to be principal
$\mathbb{P}_k^1 \times \mathbb{P}_k^1$	Remove two boundary divisors to obtain $k[s, t]$	Two independent classes $[D_1], [D_2]$	$\mathrm{Cl}(\mathbb{P}_k^1 \times \mathbb{P}_k^1) \cong \mathbb{Z}^2$
Cone $xy = z^2$	Factorial localization $k[x, x^{-1}, z]$	$\mathrm{div}(x) = 2D$	$\mathrm{Cl}(X) \cong \mathbb{Z}/2\mathbb{Z}$
Cone $xy = zw$	Factorial localization $k[x, x^{-1}, z, w]$	$\mathrm{div}(x) = D + E$	$\mathrm{Cl}(X) \cong \mathbb{Z}$
Projective morphisms	Homogeneous forms of one degree without common zeros	$\varphi^* \mathcal{O}_{\mathbb{P}^m}(1) \cong \mathcal{O}_{\mathbb{P}^n}(d)$	A nonconstant image of $\mathbb{P}^n$ has dimension n
Quasi-coherent sheaves	Modules on affine open subsets	$\widetilde{\ker \varphi} \cong \ker(\tilde{\varphi})$	Stable under kernels, images, and cokernels
Coherent sheaves	Finite modules over Noetherian rings	Finite generation is preserved under kernels and quotients	Finite geometric data remains coherent
Pullback	Tensor product $M \otimes_A B$	$g^*(\widetilde{M}) \cong \widetilde{M \otimes_A B}$	Can retain multiplicities in scheme-theoretic fibers
Pushforward	Sections or restriction of scalars	$p_* \mathcal{O}_{\mathbb{A}_k^1} \hookrightarrow k[t]$	Need not preserve coherence
Closed subschemes	Ideals and quotients A/I	$0 \longrightarrow \mathcal{I} \longrightarrow \mathcal{O}_X \longrightarrow i_* \mathcal{O}_Z \longrightarrow 0$	Remembers nilpotent and infinitesimal structure
Thickened points	$k[x]/(x^n)$	$V(x) = V(x^n)$ topologically	Same support, but different scheme structures

Table 15: Synthesis of the main constructions appearing in Problems 4–7.

10.14 Final concept map

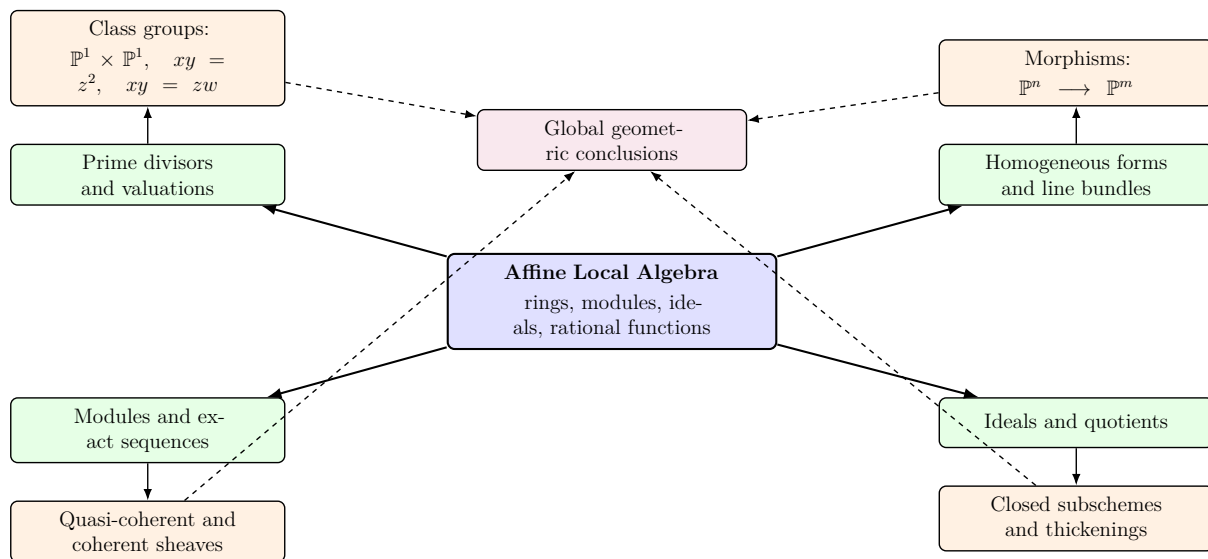


Figure 26: The unifying principle of Problems 4–7: affine algebra encodes global scheme-theoretic geometry.

The diagrams and tables above demonstrate that the four problems are not isolated calculations. Each uses the same basic translation between geometry and algebra:

prime divisors	$\longleftrightarrow$	height-one primes and valuations,
projective morphisms	$\longleftrightarrow$	homogeneous forms without base points,
quasi-coherent sheaves	$\longleftrightarrow$	modules on affine charts,
closed subschemes	$\longleftrightarrow$	quasi-coherent ideal sheaves and quotient rings.

In particular, the equations

$$\operatorname{div}(x) = 2D, \quad \operatorname{div}(x) = D+E, \quad g^*(V(t)) = V(u^2), \quad 0 \rightarrow \mathcal{I}_{Z/X} \rightarrow \mathcal{O}_X \rightarrow i_*\mathcal{O}_Z \rightarrow 0$$

are compact algebraic formulas encoding rich geometric information: torsion divisor classes, decomposed boundaries, multiplicities of fibers, and infinitesimal closed-subscheme structure.